The **irace** Package: User Guide

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1 General information

1.1 Background

The **irace** package implements an *iterated racing* procedure, which is an extension of Iterated F-race (I/F-Race) [3]. The main use of **irace** is the automatic configuration of optimization and decision algorithms, that is, finding the most appropriate settings of an algorithm given a set of instances of a problem. However, it may also be useful for configuring other types of algorithms when performance depends on the used parameter settings. It builds upon the **race** package by Birattari and it is implemented in R. The **irace** package is available from CRAN:

`https://cran.r-project.org/package=irace`

More information about **irace** is available at `https://mlopez-ibanez.github.io/irace/`.

1.2 Version

The current version of the **irace** package is 4.5.1.9000. Previous versions of the package can also be found in the CRAN website.

The algorithm underlying the current version of **irace** and its motivation are described by López-Ibáñez et al. [12]. The **adaptive capping mechanism** available from version 3.0 is described by Pérez Cáceres et al. [16]. Details of the implementation before version 2.0 can be found in a previous technical report [11].



Versions of **irace** before 2.0 are not compatible with the file formats detailed in this document.

1.3 License

The **irace** package is Copyright © 2026 and distributed under the GNU General Public License version 3.0 (`http://www.gnu.org/licenses/gpl-3.0.en.html`). The **irace** package is free software (software libre): You can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version.

The **irace** package is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Please be aware that the fact that this program is released as Free Software does not excuse you from scientific propriety, which obligates you to give appropriate credit! If you write a scientific paper describing research that made substantive use of this program, it is your obligation as a scientist to (a) mention the fashion in which this software was used in the Methods section; (b) mention the algorithm in the References section. The appropriate citation is:

> Manuel López-Ibáñez, Jérémie Dubois-Lacoste, Leslie Pérez Cáceres, Thomas Stützle, and Mauro Birattari. The **irace** package: Iterated Racing for Automatic Algorithm Configuration. *Operations Research Perspectives*, 3:43–58, 2016. doi: 10.1016/j.orp.2016.09.002

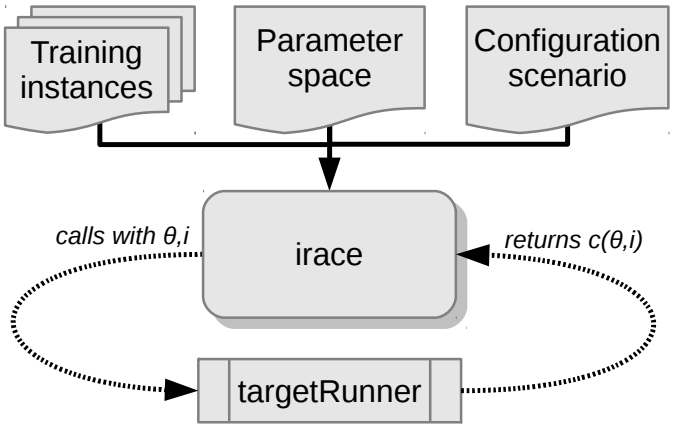


Figure 1: Scheme of **irace** flow of information.

2 Before starting

The **irace** package provides an automatic configuration tool for tuning optimization algorithms, that is, automatically finding good configurations for the parameters values of a (target) algorithm saving the effort that normally requires manual tuning.

Figure 1 gives a general scheme of how **irace** works. **Irace** receives as input a *parameter space definition* corresponding to the parameters of the target algorithm that will be tuned, a set of *instances* for which the parameters must be tuned for and a set of options for **irace** that define the *configuration scenario*. Then, **irace** searches in the parameter search space for good performing algorithm configurations by executing the target algorithm on different instances and with different parameter configurations. A `targetRunner` must be provided to execute the target algorithm with a specific parameter configuration (θ) and instance (i). The `targetRunner` function (or program) acts as an interface between the execution of the target algorithm and **irace**: It receives the instance and configuration as arguments and must return the evaluation of the execution of the target algorithm.

The following user guide contains guidelines for installing **irace**, defining configuration scenarios, and using **irace** to automatically configure your algorithms.

3 Installation

3.1 System requirements

- R (version $\geq$ 3.2.0) is required for running irace, but you don't need to know the R language to use it. R is freely available and you can download it from the R project website (`https://www.r-project.org`). See Section A for a quick installation guide of R.
- For GNU/Linux and OS X, the command-line executable `parallel-irace` requires GNU Bash. Individual examples may require additional software.

3.2 irace installation

The **irace** package can be installed automatically within R or by manual download and installation. We advise to use the automatic installation unless particular circumstances do not allow it. The

instructions to install **irace** with the two mentioned methods are the following:

3.2.1 Install automatically within R

Execute the following line in the R console to install the package:

```
install.packages("irace")
```

Select a mirror close to your location, and test the installation in the R console with:

```
library("irace")
q() # To exit R
```

Alternatively, within the R graphical interface, you may use the `Packages and data->Package installer` menu on OS X or the `Packages` menu on Windows.

3.2.2 Manual download and installation

From the **irace** package CRAN website (https://cran.r-project.org/package=irace), download one of the three versions available depending on your operating system:

- `irace_4.5.1.9000.tar.gz` (Unix/BSD/GNU/Linux)
- `irace_4.5.1.9000.tgz` (OS X)
- `irace_4.5.1.9000.zip` (Windows)

To install the package on GNU/Linux and OS X, you must execute the following command at the shell (replace `<package>` with the path to the downloaded file, either `irace_4.5.1.9000.tar.gz` or `irace_4.5.1.9000.zip`):

```
R CMD INSTALL <package>
```

To install the package on Windows, open R and execute the following line on the R console (replace `<package>` with the path to the downloaded file `irace_4.5.1.9000.zip`):

```
install.packages("<package>", repos = NULL)
```

If the previous installation instructions fail because of insufficient permissions and you do not have sufficient admin rights to install **irace** system-wide, then you need to force a local installation.

3.2.3 Local installation

Let's assume you wish to install **irace** on a path denoted by `<R_LIBS_USER>`, which is a filesystem path for which you have sufficient rights. This directory **must** exist before attempting the installation. Moreover, you must provide to R the path to this library when loading the package. However, the latter can be avoided by adding the path to the system variable `R_LIBS` or to the R internal variable `.libPaths`, as we will see below.[1]

On GNU/Linux or OS X, execute the following commands to install the package on a local directory:

[1] On Windows, see also https://cran.r-project.org/bin/windows/base/rw-FAQ.html#I-don_0027t-have-permission-to-write-to-the-R_002d3_002e3_002e1_005clibrary-directory.

```
export R_LIBS_USER="<R_LIBS_USER>"
# Create R_LIBS_USER if it doesn't exist
mkdir $R_LIBS_USER
# Replace <package> with the path to the downloaded file.
R CMD INSTALL --library=$R_LIBS_USER <package>
# Tell R where to find R_LIBS_USER
export R_LIBS=${R_LIBS_USER}:${R_LIBS}
```

On Windows, you can install the package on a local directory by executing the following lines in the R console:

```
# Replace <package> with the path to the downloaded file.
# Replace <R_LIBS_USER> with the path used for installation.
install.packages("<package>", repos = NULL, lib = "<R_LIBS_USER>")
# Tell R where to find R_LIBS_USER.
# This must be executed for every new session.
.libPaths(c("<R_LIBS_USER>", .libPaths()))
```

3.2.4 Testing the installation and invoking irace

Once **irace** has been installed, load the package and test that the installation was successful by opening an R console and executing:

```
# Load the package
library("irace")
# Obtain the installation path
system.file(package = "irace")
```

The last command must print out the filesystem path where **irace** is installed. In the remainder of this guide, the variable `$IRACE_HOME` is used to denote this path. When executing any provided command that includes the `$IRACE_HOME` variable do not forget to replace this variable with the installation path of **irace**.

On GNU/Linux or OS X, you can let the operating system know where to find **irace** by defining the `$IRACE_HOME` variable and adding it to the system `PATH`. Append the following commands to `~/.bash_profile`, `~/.bashrc` or `~/.profile`:

```
# Replace <IRACE_HOME> with the irace installation path
export IRACE_HOME=<IRACE_HOME>
export PATH=${IRACE_HOME}/bin/:$PATH
# Tell R where to find R_LIBS_USER
# Use the following line only if local installation was forced
export R_LIBS=${R_LIBS_USER}:${R_LIBS}
```

Then, open a new terminal and launch **irace** as follows:

```
irace --help
```

On Windows, you need to add both R and the installation path of **irace** to the environment variable `PATH`. To edit the `PATH`, search for "Environment variables" in the control panel, edit `PATH` and add a string similar to `C:\R_PATH\bin;C:\IRACE_HOME\bin\x64\` where `R_PATH` is the

installation path of R and `IRACE_HOME` is the installation path of **irace**. If **irace** was installed locally, you also need to edit the environment variable `R_LIBS` to add `R_LIBS_USER`. Then, open a new terminal (run program `cmd.exe`) and launch **irace** as:

```
irace.exe --help
```

Alternatively, you may directly invoke **irace** from within the R console by executing:

```
library("irace")
irace_cmdline("--help")
```

4 Running irace

Before performing the tuning of your algorithm, it is necessary to define a tuning scenario that will give **irace** all the necessary information to optimize the parameters of the algorithm. The tuning scenario is composed of the following elements:

1. Target algorithm parameter description (see Section 5.1).
2. Target algorithm runner (see Section 5.2).
3. Training instances list (see Section 5.4)
4. **irace** options (see Section 11).
5. *Optional:* Initial configurations (see Section 5.5).
6. *Optional:* Target algorithm evaluator (see Section 5.3).

These scenario elements can be provided as plain text files or as R objects. This user guide provides examples of both types, but we advise the use of plain text files, which we consider the simpler option.

For a step-by-step guide to create the scenario elements for your target algorithm continue to Section 4.1. For an example execution of **irace** using the **ACOTSP** scenario go to Section 4.2.

4.1 Step-by-step setup guide

This section provides a guide to setup a basic execution of **irace**. The template files provided in the package (`$IRACE_HOME/templates`) will be used as basis for creating your new scenario. Please follow carefully the indications provided in each step and in the template files used; if you have doubts check the the sections that describe each option in detail.

1. Create a directory (e.g., `./tuning/`) for the scenario setup. This directory will contain all the files that describe the scenario. On GNU/Linux or OS X, you can do this as follows:

   ```
   mkdir ./tuning
   cd ./tuning
   ```

2. Initialize the tuning directory with template config files. On GNU/Linux or OS X, you can do this as follows:

```
irace --init
```

3. Define the target algorithm parameters to be tuned by following the instructions in `parameters.txt`. Available parameter types and other guidelines can be found in Section 5.1.

4. *Optional*: Define the initial parameter configuration(s) of your algorithm, which allows you to provide good starting configurations (if you know some) for the tuning. Follow the instructions in `configurations.txt` and set `configurationsFile="configurations.txt"` in `scenario.txt`. More information in Section 5.5. If you do not need to define initial configurations remove this file from the directory.

5. Place the instances you would like to use for the tuning of your algorithm in the folder `./tuning/Instances/`. In addition, you can create a file (e.g., `instances-list.txt`) that specifies which instances from that directory should be run and which instance-specific parameters to use. To use such an instance file, set the appropriate option in `scenario.txt`, e.g., `trainInstancesFile = "instances-list.txt"`. See Section 5.4 for guidelines.

6. Uncomment and assign in `scenario.txt` only the options for which you need a value different from the default. Some common options that you might want to adjust are:

   `execDir` (`--exec-dir`): the directory in which **irace** will execute the target algorithm; the default value is the current directory.

   `maxExperiments` (`--max-experiments`): the maximum number of executions of the target algorithm that **irace** will perform.

   `maxTime` (`--max-time`): maximum total execution time in seconds for the executions of `targetRunner`. In this case, `targetRunner` must return two values: cost and time. Note that you must provide either `maxTime` or `maxExperiments`.

   `trainInstancesDir` (`--train-instances-dir`): set to `./Instances` if you put the training instances in that folder as instructed above.

   For setting the tuning budget, see Section 10.1. For more information on **irace** options and their default values, see Section 11.

7. Modify the `target-runner` script to run your algorithm. This script must execute your algorithm with the parameters and instance specified by **irace** and return the evaluation of the execution and *optionally* the execution time (`cost [time]`). When the `maxTime` option is used, returning `time` is mandatory. The `target-runner` template is written in GNU Bash scripting language, which can be executed easily in GNU/Linux and OS X systems. However, you may use any other programming language. We provide examples written in Python, Julia, MATLAB and other languages in `$IRACE_HOME/examples/`. Follow these instructions to adjust the given `target-runner` template to your algorithm:

   (a) Set the `EXE` variable with the path to the executable of the target algorithm.

   (b) Set the `FIXED_PARAMS` if you need extra arguments in the execution line of your algorithm. An example could be the time that your algorithm is required to run (`FIXED_PARAMS="--time 60"`) or the number of evaluations required (`FIXED_PARAMS= "--evaluations 10000"`).

   (c) The line provided in the template executes the executable described in the `EXE` variable.

   ```
   $EXE ${FIXED_PARAMS} -i ${INSTANCE} --seed ${SEED} ${CONFIG_PARAMS}
   ```

You must change this line according to the way your algorithm is executed. In this example, the algorithm receives the instance to solve with the flag `-i` and the seed of the random number generator with the flag `--seed`. The variable `CONFIG_PARAMS` adds to the command line the parameters that **irace** has given for the execution. You must set the command line execution as needed. For example, the instance might not need a flag and might need to be the first argument:

```
$EXE ${INSTANCE} ${FIXED_PARAMS} --seed ${SEED} ${CONFIG_PARAMS}
```

The output of your algorithm is saved to the file defined in the `$STDOUT` variable, and error output is saved in the file given by `$STDERR`. The line:

```
if [ -s "$STDOUT" ]; then
```

checks if the file containing the output of your algorithm is not empty. The example provided in the template assumes that your algorithm prints in the last output line the best result found (only a number). The line:

```
COST=$(cat ${STDOUT} | grep -e '^[[:space:]]*[+-]\?[0-9]' | cut -f1)
```

parses the output of your algorithm to obtain the result from the last line. The `target-runner` script must print **only** one number. In the template example, the result is printed with `echo "$COST"` (assuming `maxExperiments` is used) and the generated files are deleted (you may remove that line if you wish to keep them).



The `target-runner` script must be an executable file, unless you specify `targetRunnerLauncher`.

You can test the target runner from the R console by checking the scenario as explained earlier in Section 4.

If you have problems related to the `target-runner` script when executing **irace**, see Section B for a check list to help diagnose common problems. For more information about the `targetRunner`, please see Section 5.2,

8. *Optional*: Modify the `target-evaluator` file. This is rarely needed and the `target-runner` template does not use it. Section 5.3 explains when a `targetEvaluator` is needed and how to define it.

9. The **irace** executable provides an option (`--check`) to check that the scenario is correctly defined. We recommend to perform a check every time you create a new scenario. When performing the check, **irace** will verify that the scenario and parameter definitions are correct and will test the execution of the target algorithm. To check your scenario execute the following commands:

   - From the command-line (on Windows, execute `irace.bat`):

     ```
     # $IRACE_HOME is the installation directory of irace.
     $IRACE_HOME/bin/irace --scenario scenario.txt --check
     ```

   - Or from the R console:

     ```
     library("irace")
     scenario <- readScenario(filename = "scenario.txt",
                              scenario = defaultScenario())
     checkIraceScenario(scenario = scenario)
     ```

10. Once all the scenario elements are prepared you can execute **irace**, either using the command-line wrappers provided by the package or directly from the R console:

    - **From the command-line console**, call the command (on Windows, you should execute `irace.exe`):

      ```
      cd ./tuning/
      # $IRACE_HOME is the installation directory of irace
      # By default, irace reads scenario.txt, you can specify a different file
      # with --scenario.
      $IRACE_HOME/bin/irace
      ```

      For this example, we assume that the needed scenario files have been set properly in the `scenario.txt` file using the options described in Section 11. Most **irace** options can be specified in the command line or directly in the `scenario.txt` file.

    - **From the R console**, evaluate:

      ```
      library("irace")
      # Go to the directory containing the scenario files
      setwd("./tuning")
      scenario <- readScenario(filename = "scenario.txt",
                               scenario = defaultScenario())
      irace_main(scenario = scenario)
      ```

    This will perform one run of **irace**. See the output of `irace --help` in the command-line or `irace_cmdline("--help")` in R for quick information on additional **irace** options. For more information about **irace** options, see Section 11.



Command-line options override the same options specified in the `scenario.txt` file.

4.2 Setup example for ACOTSP

The **ACOTSP** tuning example can be found in the package installation in the folder `$IRACE_HOME/examples/acotsp`. Other example scenarios can be found in the same folder. More examples of tuning scenarios can be found in the Algorithm Configuration Library (AClib, https://www.aclib.net/).

In this section, we describe how to execute the **ACOTSP** scenario. If you wish to start setting up your own scenario, continue to the next section. For this example, we assume a GNU/Linux system such as Ubuntu with a working C compiler such as `gcc`. To execute this scenario follow these steps:

1. Create a directory for the tuning (e.g., `./tuning/`) and copy the example scenario files located in the `examples` folder to the created directory:

   ```
   mkdir ./tuning
   cd ./tuning
   # $IRACE_HOME is the installation directory of irace.
   cp $IRACE_HOME/examples/acotsp/* ./
   ls ./
   ```

```
# Make sure that target-runner is executable
chmod u+x target-runner
```

2. Download the training instances from instances.tar.gz to the ./tuning/ directory and decompress it, which creates create a folder instances:

```
tar -xvf instances.tar.gz
ls instances/
```

If the above gives an error or does not show any files, then the files were not extract correctly. Maybe the instances.tar.gz file did not download correctly or maybe it is not in the correct place. It should be within the folder tuning.correctly.

3. Download the **ACOTSP** software from https://github.com/MLopez-Ibanez/ACOTSPQAP/archive/refs/heads/master.zip to the ./tuning/ directory and compile the acotsp executable using make.

```
unzip master.zip
make -C ACOTSPQAP-master acotsp
./ACOTSPQAP-master/acotsp --help
```

If the above gives an error, then the acotsp executable failed to compile for some reason. Maybe you are missing the C compiler or some files did not extract correctly.

4. Create a directory for executing the experiments and execute **irace**:

```
mkdir ./acotsp-arena/
# $IRACE_HOME is the installation directory of irace.
$IRACE_HOME/bin/irace
```

Or you can also execute **irace** from the R console using:

```
library("irace")
setwd("./tuning/")
irace_cmdline()
```

The most usual sources of error when running the above commands are:

- The **irace** package is not correctly installed. Please make sure that installing **irace** did not give any errors.
- The location of the files is not correct. Please make sure that you have:
  - The folder tuning and that it contains the files scenario.txt, parameters-acotsp.txt, target-runner, and the folders instances, ACOTSPQAP-master and acotsp-arena.
  - The folder instances should contain the TSP instance files (*.tsp).
  - The folder ACOTSPQAP-master should contain the executable acotsp. You should be able to invoke ./ACOTSPQAP-master/acotsp --help without an error.
  - The folder acotsp-arena should be empty.

5 Defining a configuration scenario

5.1 Target algorithm parameters

The parameters of the target algorithm are defined by a parameter file as described in Section 5.1.7. Optionally, when executing **irace** from the R console, the parameters can be specified directly as an R object (see Section 5.1.8). For defining your parameters follow the guidelines provided in the following sections.

5.1.1 Parameter types

Each target parameter has an associated type that defines its domain and the way **irace** handles them internally. Understanding the nature of the domains of the target parameters is important to select appropriate types. The four basic types supported by **irace** are the following:

- *Real* parameters are numerical parameters that can take floating-point values within a given range. The range is specified as an interval '`(<lower bound>,<upper bound>)`'. This interval is closed, that is, the parameter value may eventually be one of the bounds. The possible values are rounded to a number of *decimal places* specified by the global option `digits` (Section 5.1.6). For example, given the default number of digits of 4, the values 0.12345 and 0.12341 are both rounded to 0.1234. Selected real-valued parameters can be optionally sampled on a logarithmic scale (base e).
- *Integer* parameters are numerical parameters that can take only integer values within the given range. Their range is specified as the range of real parameters and they can also be optionally sampled on a logarithmic scale (base e).
- *Categorical* parameters are defined by a set of possible values specified as '`(<value 1>, ..., <value n>)`'. The values are quoted or unquoted character strings. Empty strings and strings containing commas or spaces must be quoted.
- *Ordinal* parameters are defined by an *ordered* set of possible values in the same format as for categorical parameters. They are handled internally as integer parameters, where the integers correspond to the indexes of the values.



Boolean (or logical) parameters are best encoded as categorical ones with just two values rather than integer ones with domain $(0, 1)$. Some boolean parameters take an explicit value (0/1 or true/false) such as:

```
dlb "--dlb " c (0, 1)
```

Others are switches whose presence activates the parameter:

```
dlb "" c ("", "--dlb")
```

5.1.2 Parameter domains

For each target parameter, an interval or a set of values must be defined according to its type, as described above. There is no limit for the size of the set or the length of the interval, but keep in mind that larger domains could increase the difficulty of the tuning task. Choose always values that you consider relevant for the tuning. In case of doubt, we recommend to choose larger intervals, as occasionally best parameter settings may be not intuitive a priori. All intervals are considered as closed intervals.

It is possible to define parameters that will have always the same value. Such "*fixed*" parameters will not be tuned but their values are used when executing the target algorithm and they are

affected by constraints defined on them. All fixed parameters must be defined as categorical parameters and have a domain of one element.

5.1.3 Parameter dependent domains

Domains that are dependent on the values of other parameters can be specified only for numerical parameters (both integer and real). To do so, the dependent domain must be expressed in function of another parameter, which must be a numerical parameter. The expression that defines a dependency must be written between quotes: `(value,"expression")` or `("expression",value)` or `("expression","expression")`.

The expressions can only use the following operators and R functions: `+`, `-`, `*`, `/`, `%%`, `min`, `max`, `round`, `floor`, `ceiling`, `trunc`. If you need to use an operator or function not listed here, please contact us.



The user must ensure that the defined domain is valid at all times since **irace** currently is not able to detect possible invalid domains based on the expressions provided.

If you have a parameter `p2` that is just a transformation of another `p1`, then instead of using a dependent domain (left-hand side of the following example), it will be better to create a dummy parameter that controls the transformation (right-hand side) and do the transformation within `target-runner`. For example:

```
# With dependent domains
p1 "" r (0, 100)
p2 "" r ("p1", "p1 + 10")
```

should be

```
# With a dummy parameter
p1    "" r (0, 100)
p2dum "" r (0, 10)
```

and `target-runner` will compute $\texttt{p2} = \texttt{p2dum} \cdot \texttt{p1}$.

5.1.4 Conditional parameters

Conditional parameters are active only when others have certain values. These dependencies define a hierarchical relation between parameters. For example, the target algorithm may have a parameter `localsearch` that takes values `(sa,ts)` and another parameter `ts-length` that only needs to be set if the first parameter takes precisely the value `ts`. Thus, parameter `ts-length` is conditional on `localsearch == "ts"`.

5.1.5 Forbidden parameter configurations

A line containing just `[forbidden]` ends the list of parameters and starts the list of forbidden expressions. Each line is a logical expression (in R syntax) containing parameter names as defined by the `parameterFile` (Section 5.1), values and logical operators. For a list of R logical operators see:

https://stat.ethz.ch/R-manual/R-devel/library/base/html/Syntax.html

If **irace** generates a parameter configuration that makes any of the logical expressions evaluate to `TRUE`, then the configuration is considered forbidden and it is never evaluated. This is useful when some combination of parameter values could cause the target algorithm to crash, consume excessive CPU time or memory, or when it is known that they do no produce satisfactory results.



Initial configuration (Section 5.5) that are forbidden will be discarded with a warning.

If the forbidden constraints provided are too strict, **irace** may produce the following error:

```
irace tried 100 times to sample from the model a configuration not forbidden
without success, perhaps your constraints are too strict?
```

In that case, it may be a good idea to reformulate the forbidden constraints as conditional parameters (Section 5.1.4), parameter-dependent domains (Section 5.1.3), repairing the configurations (Section 5.6) or post-processing within the target-algorithm (Section 10.8).

5.1.6 Global options

A line containing just `[global]` starts the definition of global options. The only global option currently implemented is `digits`, which controls the number of decimal digits for real valued parameters. Its default value is 4.

5.1.7 Parameter file format

For simplicity, the description of the parameters space is given as a table. Each line of the table defines a configurable parameter

```
<name> <label> <type> <domain> [ | <condition> ]
```

where each field is defined as follows:

`<name>` The name of the parameter as an unquoted alphanumeric string, e.g., '`ants`'.

`<label>` A *label* for this parameter. This is a string that will be passed together with the parameter to `targetRunner`. In the default `targetRunner` provided with the package (Section 5.2), this is the command-line switch used to pass the value of this parameter, for instance '`"--ants "`'.
The value of the parameter is concatenated *without separator* to the label when invoking `targetRunner`, thus *any whitespace in the label is significant*. Following the same example, when parameter `ants` takes value `5`, the default targetRunner will pass the parameter as `"--ants 5"`.

`<type>` The type of the parameter, either *integer*, *real*, *ordinal* or *categorical*, given as a single letter: '`i`', '`r`', '`o`' or '`c`'. Numerical parameters can be sampled using a natural logarithmic scale with '`i,log`' and '`r,log`' (without spaces) for integer and real parameters, respectively.

`<domain>` The range or set of values of the parameter delimited by parentheses, e.g., `(0,1)` or `(a,b,c,d)`. See also parameter dependent domains (Section 5.1.3).

`<condition>` An optional *condition* that determines whether the parameter is enabled or disabled, thus making the parameter conditional. If the condition evaluates to false, then no value is assigned to this parameter, and neither the parameter value nor the corresponding label are passed to `targetRunner`. The condition must follow the same syntax as those for specifying forbidden configurations (see below), that is, it must be a valid R logical expression[2]. The condition may contain the name of other parameters as long as the dependency graph does not contain any cycle. Otherwise, **irace** will detect the cycle and stop with an error.

[2]For a list of R operators see: https://stat.ethz.ch/R-manual/R-devel/library/base/html/Syntax.html



Categorical and ordinal parameters are always treated as strings. Given a parameter like:

`a "" c (0, 5, 10, 20)`

then, a condition like `a >10` will be true when `a` is `5`, because comparisons between strings are lexicographic and `"10"` is sorted before `"5"`. As a work-around, you can convert the string to numeric in the condition with `as.numeric(a)`.

As an example, Figure 2 shows the parameters file of the **ACOTSP** scenario.

```
# name       switch             type values               [conditions (using R syntax)]
algorithm    "--"               c    (as,mmas,eas,ras,acs)
localsearch "--localsearch "  c    (0, 1, 2, 3)
alpha        "--alpha "         r    (0.00, 5.00)
beta         "--beta "          r    (0.00, 10.00)
rho          "--rho  "          r    (0.01, 1.00)
ants         "--ants "          i    (5, 100)
nnls         "--nnls "          i    (5, 50)              | localsearch %in% c(1, 2, 3)
q0           "--q0 "            r    (0.0, 1.0)           | algorithm == "acs"
dlb          "--dlb "           c    (0, 1)               | localsearch %in% c(1,2,3)
rasrank      "--rasranks "      i    (1, "ants")          | algorithm == "ras"
elitistants "--elitistants "  i    (1, 750)             | algorithm == "eas"

[forbidden]
## Examples of valid logical operators are:
## ==  !=  >=  <=  >  <  &  |  !  %in%
(alpha == 0.0) & (beta == 0.0)
```

Figure 2: Parameter file (`parameters.txt`) for tuning **ACOTSP**.

5.1.8 Parameters R format

The target parameters are stored in an R `list` that you can obtain from the R console using the following command:

```
parameters <- readParameters(file = "parameters.txt")
```

See the help of the `readParameters` function (`?readParameters`) for more information. The structure of the parameter list that is created is as follows:

`names`	Vector that contains the names of the parameters.
`types`	Vector that contains the type of each parameter 'i', 'c', 'r', 'o'.
`switches`	Vector that contains the labels of the parameters. e.g., switches to be used for the parameters on the command line.
`domain`	List of vectors, where each vector may contain two values (minimum, maximum) for real and integer parameters, or a set of values for categorical and ordinal parameters.
`conditions`	List of R logical expressions, with variables corresponding to parameter names.
`isFixed`	Logical vector that specifies which parameter is fixed and, thus, it does not need to be tuned.
`transform`	Vector that contains the transformation of each parameter. Currently, it can take values `''` (no transformation, default) of `''log''` (natural logarithmic transformation).
`nbParameters`	An integer, the total number of parameters.
`nbFixed`	An integer, the number of parameters with a fixed value.
`nbVariable`	Number of variable (i.e., to be tuned) parameters.
`isDependent`	Logical vector that specifies which parameter defines a dependent domain.
`forbidden`	List of R logical expressions that cannot evaluate to `TRUE` for any evaluated configuration.

The following example shows the structure of the `parameters` R object for the `algorithm`, `ants` and `q0` parameters of the **ACOTSP** scenario:

```
> str(parameters, vec.len = 10)

Classes 'ParameterSpace', 'R6' <ParameterSpace>
  Public:
    .params: list
    as_character: function ()
    clone: function (deep = FALSE)
    conditions: list
    depends: list
    domains: list
    forbid_configurations: function (x)
    forbidden: NULL
    get: function (x)
    get_ordered: function ()
    hierarchy: 1 1 2
    initialize: function (..., forbidden = NULL, verbose = 0L)
    isFixed: FALSE FALSE FALSE
    names: algorithm ants q0
    names_fixed:
    names_numeric: ants q0
    names_variable: algorithm ants q0
    nbFixed: 0
    nbParameters: 3
```

```
nbVariable: 3
switches: -- --ants  --q0
types: c i r
```

5.2 Target algorithm runner

The evaluation of a candidate configuration on a single instance is done by means of a user-given auxiliary program or, alternatively, a user-given R function. The function (or program name) is specified by the option `targetRunner`. The `targetRunner` must return the cost value (e.g., cost of the best solution found) of the evaluation; unless computing the cost requires information from all the configurations evaluated on an instance, e.g., when evaluating multi-objective algorithms with unknown normalisation bounds (see Section 5.3 for details).



The objective of **irace** is to minimize the cost value returned by the target algorithm. If you wish to maximize, you can multiply the cost by `-1` before returning it to **irace**.

5.2.1 Target runner as an executable program

When `targetRunner` is an auxiliary executable program, it is invoked for each candidate configuration, passing as arguments:

```
<id_configuration> <id_instance> <seed> <instance> [bound] <configuration>
```

`id_configuration`	an alphanumeric string that uniquely identifies a configuration;
`id_instance`	an alphanumeric string that uniquely identifies an instance;
`seed`	seed for the random number generator to be used for this evaluation, ignore the seed for deterministic algorithms;
`instance`	string giving the instance to be used for this evaluation;
`bound`	optional execution time bound. Only provided when the `boundMax` option is set in the scenario, see Section 10.4;
`configuration`	the pairs parameter label-value that describe this candidate configuration. Typically given as command-line switches to be passed to the executable program.

The experiment list shown in Section 5.2.2, would result in the following execution line:

```
target-runner 1 113 734718556 /home/user/instances/tsp/2000-533.tsp \
 --eas --localsearch 0 --alpha 2.92 --beta 3.06 --rho 0.6 --ants 80
```

The command line switches that describe the candidate configuration are constructed by appending to each parameter label (switch), *without separator*, the value of the parameter, following the order given in the parameter table. The program `targetRunner` must print a real number, which corresponds to the cost measure of the candidate configuration for the given instance and optionally its execution time (mandatory when `maxTime` is used and/or when the `capping` option is enabled). The working directory of `targetRunner` is set to the execution directory specified by the option `execDir`. This allows the user to execute independent runs of **irace** in parallel using different values for `execDir`, without the runs interfering with each other.

5.2.2 Target runner as an R function

When `targetRunner` is an R function, it is invoked for each candidate configuration as:

```
targetRunner(experiment, scenario)
```

where `experiment` is a list that contains information about configuration and instance to execute one experiment, and `scenario` is the scenario list. The structure of the `experiment` list is as follows:

`id_configuration`	an alphanumeric string that uniquely identifies a configuration;
`id_instance`	an alphanumeric string that uniquely identifies an instance;
`seed`	seed to be used for this evaluation;
`instance`	string giving the instance to be used for this evaluation;
`bound`	optional execution time bound;
`configuration`	1-row data frame with a column per parameter name;

The following is an example of an experiment list for the **ACOTSP** scenario:

```
> print(experiment)

$id_configuration
[1] 1

$id_instance
[1] 48

$seed
[1] 1498426593

$configuration
  algorithm localsearch alpha beta  rho ants nnls q0  dlb rasrank
1        as           0     1    1 0.95   10   NA NA <NA>      NA
  elitistants time
1          NA    5

$instance
[1] "./instances/2000-918.tsp"

$switches
       algorithm      localsearch            alpha             beta
            "--" "--localsearch "       "--alpha "        "--beta "
             rho             ants             nnls               q0
       "--rho  "        "--ants "        "--nnls "          "--q0 "
            dlb          rasrank      elitistants             time
        "--dlb "    "--rasranks " "--elitistants "        "--time "
```

If you need the `switches` of each parameter (or any other information about the parameters), you can obtain it from `scenario$parameters`.

You can find an example that calls MATLAB from R using this approach here: https://github.com/MLopez-Ibanez/irace/blob/master/inst/examples/matlabr/scenario.txt.

If targetEvaluator is NULL, then the targetRunner function must return a list with at least one element "cost", the numerical value corresponding to the evaluation of the given configuration on the given instance. A cost of Inf is accepted and results in the immediate rejection of the configuration (see Section 10.9).

If the scenario option maxTime is non-zero or if the capping option is enabled, then the list must contain at least another element "time" that reports the execution time for this call to targetRunner.

The return list may also contain the following optional elements that are used by **irace** for reporting errors in targetRunner:

error	is a string used to report an error;
outputRaw	is a string used to report the raw output of calls to an external program or function;
call	is a string used to report how targetRunner called an external program or function;

5.3 Target evaluator

Normally, targetRunner returns the cost of the execution of a candidate configuration (see Section 5.2). However, there are cases when the cost evaluation must be delayed until all candidate configurations in a race have been executed on a instance.

The targetEvaluator option defines an auxiliary program (or an R function) that allows postponing the evaluations of the candidate configurations. For each instance seen, the program targetEvaluator is only invoked after all the calls to targetRunner for all alive candidate configurations on the same instance have already finished.



When using targetEvaluator, targetRunner must not return the evaluation of the configuration. If maxTime is used, targetRunner must return only execution time.

As an example, targetEvaluator may be used to dynamically find normalization bounds for the output returned by an algorithm for each individual instance. In this case, targetRunner will save the output of the algorithm, then the first call to targetEvaluator will examine the output produced by all calls to targetRunner for the same instance, update the normalization bounds and return the normalized output. Subsequent calls to targetEvaluator for the same instance will simply return the normalized output.

A similar need arises when using quality measures for multi-objective optimization algorithms, such as the hypervolume, which typically require specifying reference points or sets. By using targetEvaluator, it is possible to dynamically compute the reference points or sets while **irace** is running. Examples are provided at examples/hypervolume. See also Section 10.2 for more information on how to tune multi-objective algorithms.

5.3.1 Target evaluator executable program

When targetEvaluator is an auxiliary executable program, it is invoked for each candidate with the following arguments:

```
<id_configuration> <id_instance> <seed> <instance> <num_configurations> <all_conf_id>
```

`id_configuration`	an alphanumeric string that uniquely identifies a configuration;
`id_instance`	an alphanumeric string that uniquely identifies an instance;
`seed`	seed to be used for this evaluation;
`instance`	string giving the instance to be used for this evaluation;
`num_configurations`	number of alive candidate configurations;
`all_conf_id`	list of IDs of the alive configurations separated by whitespace.

The `targetEvaluator` executable must print a numerical value corresponding to the cost measure of the candidate configuration on the given instance.

5.3.2 Target evaluator R function

When `targetEvaluator` is an R function, it is invoked for each candidate configuration as:

```
targetEvaluator(experiment, num_configurations, all_conf_id, scenario,
                target_runner_call)
```

where `experiment` is a list that contains information about one experiment (see Section 5.2.2), `num_configurations` is the number of configurations alive in the race, `all_conf_id` is the vector of IDs of the alive configurations, `scenario` is the scenario list and `target_runner_call` is the string of the `targetRunner` execution line.

The function `targetEvaluator` must return a list with one element `"cost"`, the numerical value corresponding to the cost measure of the given configuration on the given instance.

The return list may also contain the following optional elements that are used by **irace** for reporting errors in `targetEvaluator`:

`error`	is a string used to report an error;
`outputRaw`	is a string used to report the raw output of calls to an external program or function;
`call`	is a string used to report how `targetEvaluator` called an external program or function;

5.4 Training instances

The **irace** options `trainInstancesDir` and `trainInstancesFile` specify where to find the training instances. If you only set the value of `trainInstancesDir` (e.g., to `./Instances`), **irace** will consider all files within that directory as training instances.

Otherwise, the value of `trainInstancesFile` may specify a text file. The format of this file is one instance per line. Within each line, elements separated by white-space will be parsed as separate arguments to be supplied to `targetRunner`. This allows defining instance-specific parameter settings. Quoted strings will be parsed as a single argument. The following example shows a training instance file for the **ACOTSP** scenario:

```
# Example training instances file
100/100-1_100-2.tsp --time 1
100/100-1_100-3.tsp --time 2
100/100-1_100-4.tsp --time 3
```

Figure 3: Training instances file for tuning **ACOTSP**.

The value of `trainInstancesDir`, if set, is always prefixed to the instance name, that is, the instances names are treated as relative to this directory. For example, given the above file as `trainInstancesFile` and `trainInstancesDir="./Instances"`, then a possible invocation of `targetRunner` would be:

```
target-runner 1 4 5718 ./Instances/100/100-1_100-2.tsp --time 1 --alpha 2.92 ...
```

Training instances do not need to be files, **irace** just passes the elements of each line as arguments to `targetRunner`, thus each line may denote the name of a benchmark function or a label, plus instance-specific settings, that the target algorithm understands. Each line may even be the command-line parameters required to call an instance generator within `targetRunner`. When the instances do not represent actual files, then `trainInstancesDir` is usually set to the empty string (`--train-instances-dir=""`). For example,

```
# Example training instances file
rosenbrock_20 --function=12 --nvar 20
rosenbrock_30 --function=12 --nvar 30
rastrigin_20 --function=15 --nvar 20
rastrigin_30 --function=15 --nvar 30
```

Optionally, when executing **irace** from the R console, the list of instances might be provided explicitly by means of the variable `scenario$instances`. Thus, the previous example would be equivalent to:

```
scenario$instances <- c("rosenbrock_20 --function=12 --nvar 20",
                        "rosenbrock_40 --function=12 --nvar 30",
                        "rastrigin_20 --function=15 --nvar 20",
                        "rastrigin_40 --function=15 --nvar 30")
```

By default, **irace** assumes that the target algorithm is stochastic (the value of the option `deterministic` is 0), thus, the same configuration can be executed more than once on the same instance and obtain different results. In this case, **irace** generates pairs `(instance,seed)` by generating a random seed for each instance. In other words, configurations evaluated on the same instance use the same random seed. This is a well-known variance reduction technique called *common random numbers* [15]. If all available pairs are used within a run of **irace**, new pairs are generated with different seeds, that is, a configuration evaluated more than once per instance will use different random seeds.

If `deterministic` is set to 1, then each instance will be used at most once per race. This setting should only be used for target algorithms that do not have a stochastic behavior and, therefore, executing the target algorithm on the same instance several times with different seeds does not make sense.



If `deterministic` is active and the number of training instances provided to **irace** is less than `firstTest` (default: 5), no statistical test will be performed on the race.

Finally, **irace** randomly re-orders the sequence of instances provided. This random sampling may be disabled by using the option `sampleInstances` (`--sample-instances 0`) if keeping the order provided in the instance file is important.



We advise to always sample instances to prevent biasing the tuning due to the instance order. See also Section 10.6

5.5 Initial configurations

The scenario option `configurationsFile` allows specifying a text file that contains an initial set of configurations to start the execution of **irace**. If the number of initial configurations supplied in the file is less than the number of configurations required by **irace** in the first iteration, additional configurations will be sampled uniformly at random.

The format of the configurations file is one configuration per line, and one parameter value per column. The first line must give the parameter name corresponding to each column (names must match those given in the parameters file). Each configuration must satisfy the parameter conditions (`NA` should be used for those parameters that are not enabled for a given configuration) and not be forbidden by the constraints that define forbidden configurations (Section 5.1.5), if any.

Figure 4 gives an example file that corresponds to the **ACOTSP** scenario.

```
## Initial candidate configuration for irace
algorithm localsearch alpha beta rho  ants nnls dlb q0 rasrank elitistants
as        0           1.0   1.0  0.95 10   NA   NA  0  NA      NA
```

Figure 4: Initial configuration file (`default.txt`) for tuning **ACOTSP**.

We advise to use this feature when a default configuration of the target algorithm exists or when different sets of good parameter values are known. This will allow **irace** to start the search from those parameter values and attempt to improve their performance.

5.6 Repairing configurations

In some problems, the parameter values require complex constraints that cannot be implemented by constraints defined in the parameter space (Section 5.1.5). The scenario option `repairConfiguration` can be set to a user-defined R function that takes a single configuration generated by irace and returns a "*repaired*" configuration, thus allowing the implementation of any rules necessary to satisfy arbitrary constraints on parameter values. The `repairConfiguration` function is called after generating a configuration and before checking for forbidden configurations. The first argument is a 1-row `data.frame` with parameter names as the column names and the second argument is the `parameters` list (Section 5.1.8). An example that makes all real-valued parameters sum up to one would be:

```
repairConfiguration = function(configuration, parameters)
{
  isreal <- names(which(parameters$types[colnames(configuration)] == "r"))
  # This ignores 'digits'
  c_real <- unlist(configuration[isreal])
  c_real <- c_real / sum(c_real)
  configuration[isreal] <- c_real
  return(configuration)
}
```

The following example forces three specific parameters to be in increasing order:

```
repairConfiguration = function(configuration, parameters)
{
 columns <- c("p1","p2","p3")
```

```
 # cat("Before"); print(configuration)
 configuration[columns] <- sort(unlist(configuration[columns], use.names=FALSE))
 # cat("After"); print(configuration)
 return(configuration)
}
```

The above code can be specified directly in the `scenarioFile`, by default `scenario.txt`.

6 Parallelization

A single run of **irace** can be done much faster by executing the calls to `targetRunner` (the runs of the target algorithm) in parallel. There are four ways to parallelize a single run of **irace**:

1. **Parallel processes**: The option `parallel` executes multiple calls to `targetRunner` in parallel within a single computer, by means of the **parallel** R package. For example, adding `--parallel N` to the command line of **irace** will launch in parallel up to N calls of the target algorithm. When using this option within a computing cluster, **irace** will be submitted as a *job* in some way that tells the cluster to "reserve" N CPUs (or tasks depending on the cluster) within a single cluster node (a single machine).

2. **MPI**: By enabling the option `mpi`, calls to `targetRunner` will be executed in parallel by using the message passing interface (MPI) protocol (requires the **Rmpi** R package). In this case, the option `parallel` controls the number of slave nodes used by **irace**. For example, adding `--mpi 1 --parallel N` to the command-line will create N slaves + 1 master, and execute up to N calls of `targetRunner` in parallel.

   The user is responsible for setting up the required MPI environment. MPI is commonly available in computing clusters and requires launching **irace** in some particular way. An example script for using MPI mode in a SGE cluster is given at `$IRACE_HOME/bin/parallel-irace-mpi`.

   By default, **irace** dynamically balances the load among nodes, however, this may significantly increase communication overhead in some parallel environments and disabling `loadBalancing` may be faster.

3. **Batch queue mode**: Some computing clusters work by submitting jobs to a batch queue and waiting for the jobs to finish. With the option `batchmode` (`--batchmode [sge|pbs|torque|slurm]`), **irace** will launch in parallel as many calls of `targetRunner` as possible and use a cluster-specific method to wait for jobs to finish. In this mode, `parallel` controls how many jobs are queued and should be set to the queue limit of your cluster. If your cluster type is not supported or not working as expected, please submit a pull request (https://github.com/MLopez-Ibanez/irace/pulls) adding support to your cluster type. See the examples in `$IRACE_HOME/examples/batchmode-cluster/`.



In batchmode, **irace** runs in the submission node of the cluster, hence, **irace** is not submitted to the cluster as a job (that is, neither `qsub` nor `squeue` should be used to invoke **irace** itself). The user must call the appropriate job submission command (e.g., `qsub` or `squeue`) from `targetRunner` with the appropriate settings for their cluster, that is, `targetRunner` submits one job to the cluster and prints a single string: The job ID that allows **irace** to determine the status of the running job. Moreover, the use of a separate `targetEvaluator` script is required to collect the results of `targetRunner` and return them to **irace**.

4. `targetRunnerParallel`: This option allows users to fully control the parallelization of the execution of `targetRunner`. Its value must be an R function that will be invoked by **irace** as follows:

```
targetRunnerParallel(experiments, exec_target_runner, scenario, target_runner)
```

where `scenario` is the list describing the configuration scenario (Section 5); `experiments` is a list that describes the configurations and instances to be executed (see Section 5.2 for a description); `target_runner` is the function that calls the target algorithm and it is the same as `targetRunner`, if the latter is a function, or it is a call to `target_runner_default`, if `targetRunner` is the path to an executable; and `exec_target_runner` is an internal function within **irace** that takes care of executing `target_runner`, check its output and, possibly, retry in case of error (see `targetRunnerRetries`). The `targetRunnerParallel` function should call the given `target_runner` function for each element in the `experiments` list, possibly using `exec_target_runner` as a wrapper. A trivial example would be:

```
targetRunnerParallel <- function(experiments, exec_target_runner, scenario,
                                 target_runner)
{
  lapply(experiments, exec_target_runner, scenario = scenario,
         target_runner = target_runner)
}
```

However, the user is free to set up the calls in any way, perhaps implementing their own replacement for `target_runner` and/or `exec_target_runner`. The user may load and call other R packages, such as **batchtools** (https://cran.r-project.org/package=batchtools).

The only requirement is that the `targetRunnerParallel` function must return a list of the same length as `experiments`, where each element is the output expected from the corresponding call to `targetRunner` (see Section 5.2).

The following is an example of the output of a call to `targetRunnerParallel` with 2 experiments, in which the execution time is not reported:

```
   print(output)

## [[1]]
## [[1]]$cost
## [1] 42220233
##
## [[1]]$time
## numeric(0)
##
##
## [[2]]
## [[2]]$cost
## [1] 34472321
##
## [[2]]$time
## numeric(0)
```

The best option will depend on the resources available to you. Option 1 is usually the fastest and simplest to setup. Running on a node (machine) with 128 CPUs will be faster than running on 8 nodes with 16 CPUs because the communication between nodes required by MPI can be very slow depending on the cluster. Option 2 may be faster if irace generates more configurations per iteration than the number of CPUs of a single node. However, depending on the configuration of your cluster, requesting many CPUs may require waiting in the queue a long time. Option 3 may be the slowest since irace has to check the queue frequently. However, irace will start running experiments as soon as 1 CPU is available, thus option 3 may actually finish earlier than the other options if there is always some CPUs available in the cluster but the queue for requesting many CPUs at once is very long.

As a rule-of-thumb, if you only have access to a single machine, then you only need option 1. If you have access to a computing cluster with multiple machines, then use option 1 with the maximum number of CPUs that a single node has in your computing cluster. If that number is 64 or more, it should be enough unless a single run of irace evaluates thousands of configurations. Otherwise, investigate option 2. If option 2 does not work, then investigate option 3.

7 Testing (Validation) of configurations

Once the tuning process is finished, **irace** returns a set of configurations corresponding to the elite configurations at the end of the run, ordered from best to worst. In order to evaluate the generality of these configurations without looking at their performance on the training set, **irace** offers the possibility of evaluating these configurations on a test instance set, typically different from the training set used during the tuning phase. These evaluations will use the same settings for parallel execution, `targetRunner` and `targetEvaluator`.

The test instances can be specified by the options `testInstancesDir` and/or `testInstancesFile`, or by setting directly the variable `scenario$testInstances`. These options behave similarly to their counterparts for the training instances (Section 5.4). In particular, each test instance is assigned a different seed in the same way as done for the training instances. In principle, **irace** evaluates each configuration on each testing instance just once, because evaluating one run on n instances is always better than evaluating n' runs on n/n' instances [2]. However, if the number of instances is limited, one can always duplicate instances as needed in the `testInstancesFile`, and **irace** will assign a different random seed to each instance. An example of the output produced by **irace** when testing is shown in Fig. 5.

The options `testNbElites` and `testIterationElites` control which configurations are evaluated during the testing phase. In particular, setting `testIterationElites = 1` will test not only the final set of elite configurations (those returned at the end of the training phase), but also the set of elites at the end of each race (iteration). The option `testNbElites` limits the maximum number of configurations considered within each set. Some examples:

- `testIterationElites = 0; testNbElites = 1` means that only the best configuration found during the run of **irace**, the final best, will be used in the testing phase.
- `testIterationElites = 1; testNbElites = 1` will test, in addition to the final best, the best configuration found at each iteration.
- `testIterationElites = 1; testNbElites = 2` will test the two best configurations found at each iteration, in addition to the final best and second-best configurations.

The testing can be also (re-)executed at a later time by using the following R command (but you may need to override `testNbElites` and `testIterationElites`):

```
testing_fromlog(logFile = "./irace.Rdata", testNbElites = 1)
```

The above line will load the scenario setup from `logFile` to perform the testing. The testing results will be stored in the R object `iraceResults$testing`, which is saved in the file specified by `scenario$logFile`. The structure of the object is described in Section 9.2. For examples on how to analyse the results see Section 9.3.

Another alternative is to test a specific set of configurations using the command-line option `--only-test` as follows:

```
irace --only-test configurations.txt
```

where `configurations.txt` has the same format as the set of initial configurations (Section 5.5).

8 Recovering irace runs

Problems like power cuts, hardware malfunction or the need to use computational power for other tasks may occur during the execution of **irace**, terminating a run before completion. At the end of each iteration, **irace** saves an R data file (`logFile`, by default `"./irace.Rdata"`) that not only contains information about the tuning progress (Section 9.2), but also internal information that allows recovering an incomplete execution.

To recover an incomplete **irace** run, set the option `recoveryFile` to the log file previously produced, and **irace** will continue the execution from the last saved iteration. The state of the random generator is saved and loaded, therefore, as long as the execution is continued in the same machine, the obtained results will be exactly the same as executing **irace** in one step (external factors, such as CPU load and disk caches, may affect the target algorithm and that may affect the results). You can specify the `recoveryFile` from the command-line or from the scenario file, and execute **irace** as described in Section 4. For example, from the command-line use:

```
irace --recovery-file "./irace-backup.Rdata"
```



When recovering a previous run, **irace** will try to save data on the file specified by the `logFile` option. Thus, you must specify different files for `logFile` and `recoveryFile`. Before recovering, we strongly advise to rename the saved R data file as in the example above, which uses `"irace-backup.Rdata"`.



Do not change anything in the log file or the scenario file before recovering, as it may have unexpected effects on the recovered run of **irace**. In case of doubt, please contact us first (Section 13). In particular, it is not possible to continue a run of **irace** by recovering with a larger budget. Results will **not** be the same as running **irace** from the start with the largest budget. An alternative is to use the final configurations from one run as the initial configurations of a new run.



If your scenario uses `targetEvaluator` (Section 5.3) and `targetEvaluator` requires files created by `targetRunner`, then recovery will fail if those files are not present in the `execDir` directory. This can happen, for example, if you recover from a different directory than the one from which irace was initially executed, or when `execDir` is set to a temporary directory for every irace run. Thus, you need to copy the contents of the previous `execDir` into the new one.

```
...

# Testing of elite configurations: 5
# Testing iteration configurations: TRUE
# 2023-10-01 13:35:17 BST: Testing configurations (in no particular order): 2 29 3 20 50 34 47 86 74 111 106 92 123 134 130 119
    algorithm localsearch  alpha   beta    rho ants nnls     q0 dlb rasrank elitistants
2         acs           3 1.1275 3.3469 0.6471   36   43 0.9053   0      NA          NA
29        ras           3 4.5152 5.6811 0.7818   11   43     NA   0      13          NA
3         eas           3 3.7246 5.5153 0.5998   75   29     NA   1      NA          91
20        acs           2 1.4734 0.1808 0.4304    5   39 0.5665   1      NA          NA
50        ras           3 2.2804 4.5294 0.5508   18   32     NA   1      12          NA
34        acs           3 2.2044 2.4923 0.7243   12   29 0.4628   0      NA          NA
47        acs           2 1.7046 6.3908 0.3256    5   36 0.2288   0      NA          NA
86        ras           3 3.9808 3.4401 0.2191   20   25     NA   1      15          NA
74       mmas           3 1.1135 0.6356 0.3765   11   33     NA   1      NA          NA
111       acs           2 1.2881 6.4311 0.5859   10   23 0.2126   0      NA          NA
106       ras           3 3.2756 5.5035 0.9277   18   32     NA   0      10          NA
92        acs           2 1.6234 9.2153 0.1101    8   40 0.3412   0      NA          NA
123       acs           2 1.0584 5.8121 0.5587   10   35 0.2575   0      NA          NA
134       acs           2 1.6144 7.8972 0.1999    8   33 0.1415   1      NA          NA
130       acs           2 1.5899 6.9391 0.5529    8   26 0.3796   0      NA          NA
119       acs           2 1.4432 6.4746 0.4582    5   28 0.1931   0      NA          NA
# 2023-10-01 13:48:55 BST: Testing results (column number is configuration ID in no particular order):
         seeds        2       29        3       20       50       34       47       86       74       111       106       92       123
1t  1385446146 23609115 23432130 23400390 23403445 23383391 23439897 23380458 23455774 23355888 23382496 23417159 23404000 23492199 2342
2t  1396979195 23327035 23208321 23186931 23288822 23181225 23228238 23265383 23201401 23279356 23188848 23171998 23314132 23244891 2320
3t   448912041 23111373 23246027 23092265 23130590 23016527 23056027 23070519 23084400 23041364 23118905 23063197 23115280 23069064 2309
4t   732530909 23128507 23159085 23119773 23290945 23063257 23095963 23110872 23151621 23053285 23133797 23139894 23148536 23101345 2308
5t   947545849 23340533 23347633 23298513 23274844 23236071 23274590 23246473 23292452 23232446 23242908 23264111 23219377 23258853 2323
6t   195435663 23532942 23470894 23496016 23533573 23423201 23538541 23526779 23500842 23497417 23525281 23464461 23510384 23454321 2347
7t   798649446 23440321 23430267 23331305 23422861 23375048 23454558 23491515 23405251 23509688 23386282 23508539 23438775 23432504 2343
8t  1692971486 23376183 23262794 23300201 23286312 23258955 23368034 23301552 23263607 23231022 23254721 23271971 23236607 23227995 2328
9t   896414478 23375100 23296796 23375905 23293303 23293631 23428815 23292599 23376861 23314826 23312995 23369543 23333975 23355279 2335
10t 1371337352 23222214 23163714 23110039 23126033 23089358 23198436 23080692 23146608 23106011 23092382 23087101 23108507 23096788 2312
# 2023-10-01 13:48:55 BST: Finished testing
```

Figure 5: Sample text output of **irace** when evaluating on test instances.

9 Output and results

During its execution, **irace** prints information about the progress of the tuning in the standard output. Additionally, after each iteration, an R data file is saved (`logFile` option) containing the state of **irace**.

9.1 Text output

Figure 6 shows the output, up to the end of the first iteration, of a run of elitist **irace** applied to the **ACOTSP** scenario with 1000 evaluations as budget.

First, **irace** gives the user a warning informing that it has found a file with the default scenario filename and it will use it. Then, general information about the selected **irace** options is printed:

- `nbIterations` indicates the minimum number of iterations **irace** has calculated for the scenario. Depending on the development of the tuning the final iterations that are executed can be more.
- `minNbSurvival` indicates the minimum number of alive configurations that are required to continue a race. When less configurations are alive the race is stopped and a new iteration begins.
- `nbParameters` is the number of parameters of the scenario.
- `seed` is the number that was used to initialize the random number generator in **irace**.
- `confidence level` is the confidence level of the statistical test.
- `budget` is the total number of evaluations available for the tuning.
- `time budget` is the maximum execution time available for the tuning.
- `mu` is a value used for calculating the minimum number of iterations.
- `deterministic` indicates if the target algorithm is assumed to be deterministic.

At each iteration, information about the progress of the execution is printed as follows:

- `experimentsUsed` is the number of experiments from the total budget that have been used up to the current iteration.
- `timeUsed` is the execution time used so far in the experiments. Only available when reported in the `targetRunner` (activate it with the `maxTime` option).
- `remainingBudget` is the number of experiments that have not been used yet.
- `boundEstimate` estimation of the mean execution time. This is used to calculate the remaining budget when `maxTime` is used.
- `currentBudget` is the number of evaluations **irace** has allocated to the current iteration.
- `nbConfigurations` is the number of configurations **irace** will use in the current iteration. In the first iteration, this number of configurations include the initial configurations provided; in later iterations, it includes the elite configurations from the previous iterations.

After the iteration information, a table shows the progress of the iteration execution. Each row of the table gives information about the execution of an instance in the race. The first column contains a symbol that describes the results of the statistical test:

`|x|` No statistical test was performed for this instance. The options `firstTest` and `eachTest` control on which instances the statistical test is performed.

`|-|` Statistical test performed and configurations have been discarded. The column `Alive` gives an indication of how many configurations have been discarded.

`|=|` Statistical test performed and no configurations have been discarded. This means **irace** needs to evaluate more instances to identify the best configurations.

`|!|` This indicator exists only for the elitist version of **irace**. It indicates that the statistical test was performed and some elite configurations appear to show bad performance and could be discarded but they are kept because of the elitist rules. See option `elitist` in Section 11 for more information.

Other columns have the following meaning:

`Instance`: Index of `(instance,seed)` pair executed. This number corresponds to the row in the data frame returned by `get_instanceID_seed_pairs()`. See Section 9.2 for more information. This index is different from the instance ID passed to `targetRunner`.

`Bound`: Only when `capping` is enabled. Execution time used as bound for the execution of new candidate configurations.

`Alive`: Number of configurations that have not been discarded after the statistical test was performed.

`Best`: ID of the best configuration according to the instances seen so far in this race (i.e., not including previous iterations).

`Mean best`: Mean cost value of the best configuration across the instances seen so far in this race (not globally). Equivalent to the concept of "iteration-best" in other algorithms.

`Exp so far`: Number of experiments performed so far.

`W time`: Wall-clock time spent on this instance.

`rho`, `KenW`, and `Qvar`: Spearman's rank correlation coefficient rho, Kendall's concordance coefficient W, and a variance measure described in [17], respectively, of the configurations across the instances evaluated so far in this iteration. These measures evaluate how consistent is the performance of the configurations across the instances. Values close to 1 for `rho` and `KenW` and values close to 0 for `Qvar` indicate that the scenario is highly homogeneous. For heterogeneous scenarios, we provide advice in Section 10.6.

Finally, **irace** outputs the best configuration found and a list of the elite configurations. The elite configurations are configurations that did not show statistically significant difference during the race; they are ordered according to their mean performance on the executed instances.

```
#-------------------------------------------------------------------------------
# irace: An implementation in R of (Elitist) Iterated Racing
# Version: 4.4.4.ae27aa1
# Copyright (C) 2010-2025
# Manuel Lopez-Ibanez     <manuel.lopez-ibanez@manchester.ac.uk>
# Jeremie Dubois-Lacoste
# Leslie Perez Caceres    <leslie.perez.caceres@ulb.ac.be>
#
# This is free software, and you are welcome to redistribute it under certain
# conditions.  See the GNU General Public License for details. There is NO
# WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
#
# irace builds upon previous code from the race package:
#     race: Racing methods for the selection of the best
#     Copyright (C) 2003 Mauro Birattari
#-------------------------------------------------------------------------------
# installed at: /home/manu/R/x86_64-pc-linux-gnu-library/4.3/irace
# called with: --parallel 2
# 2026-09-02 08:25:10 BST: Reading parameter file '/home/manu/work/irace/git/devel-examples/vignette-example/parameters.txt'.
# 2026-09-02 08:25:10 BST: 1 expression(s) specifying forbidden configurations read.
# 2026-09-02 08:25:10 BST: Read 1 configuration(s) from file '/home/manu/work/irace/git/devel-examples/vignette-example/default.txt'
# 2026-09-02 08:25:10 BST: Initialization
# Elitist race
# Elitist new instances: 1
# Elitist limit: 2
# nbIterations: 5
# minNbSurvival: 5
# nbParameters: 11
# seed: 687542627
# confidence level: 0.95
# budget: 1000
# mu: 5
# deterministic: FALSE

# 2026-09-02 08:25:10 BST: Iteration 1 of 5
# experimentsUsed: 0
# remainingBudget: 1000
# currentBudget: 200
# nbConfigurations: 33
# Markers:
     x No test is performed.
     c Configurations are discarded only due to capping.
     - The test is performed and some configurations are discarded.
     = The test is performed but no configuration is discarded.
     ! The test is performed and configurations could be discarded but elite configurations are preserved.
     . Alive configurations were already evaluated on this instance and nothing is discarded.
     : All alive configurations are elite, but some need to be evaluated on this instance.

+-+-----------+-----------+-----------+---------------+-----------+--------+-----+----+------+
| |   Instance|      Alive|       Best|       Mean best| Exp so far|  W time|  rho|KenW|  Qvar|
+-+-----------+-----------+-----------+---------------+-----------+--------+-----+----+------+
|x|          1|         33|         31|     33419384.00|         33|00:01:35|   NA|  NA|    NA|
|x|          2|         33|         31|     33024254.00|         66|00:01:33|+0.94|0.97|0.0115|
|x|          3|         33|         31|     33114822.33|         99|00:01:33|+0.96|0.97|0.0079|
|x|          4|         33|         31|     33049172.00|        132|00:01:33|+0.96|0.97|0.0070|
|-|          5|          2|         31|     33017978.60|        165|00:01:32|+0.20|0.36|0.4000|
+-+-----------+-----------+-----------+---------------+-----------+--------+-----+----+------+
Best-so-far configuration:          31    mean value:      33017978.60
Description of the best-so-far configuration:
   .ID. algorithm localsearch  alpha   beta    rho ants nnls    q0 dlb rasrank elitistants time .PARENT.
31   31       acs           3 4.2717 0.2871 0.9362    6   17 0.918   1      NA          NA    5       NA

# 2026-09-02 08:32:59 BST: Elite configurations (first number is the configuration ID; listed from best to worst according to the sum of
   algorithm localsearch  alpha   beta    rho ants nnls    q0 dlb rasrank elitistants time
31       acs           3 4.2717 0.2871 0.9362    6   17 0.918   1      NA          NA    5
9       mmas           3 3.4904 4.9746 0.5959   13   33    NA   0      NA          NA    5
# 2026-09-02 08:32:59 BST: Iteration 2 of 5
# experimentsUsed: 165
```

Figure 6: Sample text output of **irace**.

9.2 R data file (logFile)

The R data file created by **irace** (by default as `irace.Rdata`, see option `logFile`) contains an object called `iraceResults`. You can load this file in the R console with:

```
logfile <- system.file(package="irace", "exdata", "irace-acotsp.Rdata", mustWork=TRUE)
iraceResults <- read_logfile(logfile)
```

The `iraceResults` object is a list, and the elements of a list can be accessed in R by using the `$` or `[[]]` operators:

```
> iraceResults$irace_version

[1] "4.4.4.ae27aa1"
```

The `iraceResults` list contains the following elements:

- `scenario`: The scenario R object containing the **irace** options used for the execution. See Section 11 and the help of the **irace** package; open an R console and type: `?defaultScenario`. See Section 11 for more information.
- `parameters`: The parameters R object containing the description of the target algorithm parameters. See Section 5.1.
- `allConfigurations`: The target algorithm configurations generated by **irace**. This object is a `data frame`, each row is a candidate configuration; the first column (`.ID.`) indicates the internal identifier of the configuration; the final column (`.PARENT.`) is the identifier of the configuration from which the current configuration was sampled; and the remaining columns correspond to the parameter values; each column is named as the parameter name specified in the parameter object.

```
> head(iraceResults$allConfigurations)

  .ID. algorithm localsearch  alpha   beta    rho ants nnls     q0
1    1        as           0 1.0000 1.0000 0.9500   10   NA     NA
2    2      mmas           2 0.3654 6.2246 0.9672   19   50     NA
3    3       ras           0 2.8654 1.2246 0.4722   86   NA     NA
4    4       acs           3 1.6154 8.7246 0.2247   41   39 0.7617
5    5       eas           1 4.1154 3.7246 0.7197    9   16     NA
6    6        as           0 2.2404 7.4746 0.8434    6   NA     NA
   dlb rasrank elitistants time .PARENT.
1 <NA>      NA          NA    5       NA
2    1      NA          NA    5       NA
3 <NA>      42          NA    5       NA
4    0      NA          NA    5       NA
5    1      NA         602    5       NA
6 <NA>      NA          NA    5       NA
```

- `allElites`: A list that contains one element per iteration. Each element contains the internal identifier of the elite candidate configurations of the corresponding iteration (identifiers correspond to `allConfigurations$.ID.`).

```
> print(iraceResults$allElites)

[[1]]
[1] 31  9

[[2]]
[1] 31  9 49

[[3]]
[1] 31  9 49 72

[[4]]
[1]  98  31  96  49 103

[[5]]
[1] 120 136 150 134  98

[[6]]
[1] 136 150 120 166 134

[[7]]
[1] 167 136 150 120 166

[[8]]
[1] 167 120 136 150 166
```

The configurations are ordered by mean performance, that is, the ID of the best configuration corresponds to the first ID. To obtain the values of the parameters of all elite configurations found by **irace** use:

```
> logfile <- system.file(package="irace", "exdata", "irace-acotsp.Rdata", mustWork=TRUE)
> getFinalElites(logfile, n = 0)

    .ID. algorithm localsearch  alpha   beta    rho ants nnls q0 dlb
120  120        as           3 3.9042 2.3662 0.8216   13   16 NA   1
136  136        as           3 3.9196 0.3768 0.9887    8   22 NA   1
150  150        as           3 3.7455 2.5667 0.7661   15   15 NA   1
166  166        as           3 4.0313 1.4703 0.8952   20   13 NA   1
167  167        as           3 4.0264 1.6084 0.9609   13   17 NA   1
    rasrank elitistants time .PARENT.
120      NA          NA    5       98
136      NA          NA    5       98
150      NA          NA    5       98
166      NA          NA    5      120
167      NA          NA    5      150
```

- `iterationElites`: A vector containing the best candidate configuration ID of each iteration. The best configuration found corresponds to the last one of this vector.

```
> print(iraceResults$iterationElites)

[1]  31  31  31  98 120 136 167 167
```

One can obtain the full configuration with:

```
> last <- length(iraceResults$iterationElites)
> id <- iraceResults$iterationElites[last]
> getConfigurationById(iraceResults, ids = id)

    .ID. algorithm localsearch  alpha   beta    rho ants nnls q0 dlb
167  167        as           3 4.0264 1.6084 0.9609   13   17 NA   1
    rasrank elitistants time .PARENT.
167      NA          NA    5      150
```

- `rejectedConfigurations`: A vector containing the rejected configurations IDs. These correspond to configurations that produced failed executions and were ignored by **irace** during the configuration process. See Section 10.9 to enable the detection of such configurations.
- `experiments`: A matrix with configurations as columns and instances as rows. Column names correspond to the internal identifier of the configuration (`allConfigurations$.ID.`). The results of a particular configuration can be obtained using:

```
> # As an example, we use the best configuration found
> best_config <- getFinalElites(iraceResults, n = 1)
> best_id <- as.character(best_config$.ID.)
> # Obtain the results of the best configuration
> all_exp <- iraceResults$experiments[, best_id]
> # all_exp is a vector and names(all_exp) is the (instance,seed) index.
> all_exp

       1        2        3        4        5        6        7
33343832 32695592 33197088 32853081 32966323 32845751 32874384
       8        9       10       11       12       13       14
32887135 32822326 32959516 32737406 32671625 32789499 32573629
      15
32899982

> # Obtain the results of the first and best configurations
> all_exp <- iraceResults$experiments[, c("1", best_id)]
> # all_exp is a matrix: colnames(all_exp) is configurationID and
> # rownames(all_exp) is the (instance,seed) index.
> all_exp

          1      167
1  42220233 33343832
2  39533155 32695592
3  41303115 33197088
```

```
4  40251195 32853081
5  39741584 32966323
6        NA 32845751
7        NA 32874384
8        NA 32887135
9        NA 32822326
10       NA 32959516
11       NA 32737406
12       NA 32671625
13       NA 32789499
14       NA 32573629
15       NA 32899982
```

When a configuration was not executed on an instance, its value is `NA`. A configuration may not be executed on an instance because: (1) it was not created yet when the instance was used, or (2) it was discarded by the statistical test and not executed on subsequent instances, or (3) the race terminated before this instance was considered.

Row names correspond to the row index of the `(instance,seed)` pairs in the data frame returned by `get_instanceID_seed_pairs()`. The instanceID and seed used for a particular experiment can be obtained with:

```
> # As an example, we get instanceID, seeds and instances of the experiments
> # of the best configuration.
> # We could get the indexes of the instances on which at least one
> # configuration was executed:
> pair_index <- which(apply(!is.na(all_exp), 1L, any))
> # or the instances on which all configurations were executed:
> pair_index <- which(apply(!is.na(all_exp), 1L, all))
> # but in this example we get the indexes of the instances executed for
> # the best configuration.
> pair_index <- which(!is.na(all_exp[, best_id]))
> instanceID <- get_instanceID_seed_pairs(iraceResults)[["instanceID"]][pair_index]
> # or get the seeds
> get_instanceID_seed_pairs(iraceResults)[["seed"]][pair_index]

 [1] 1498426593 1324006684  156117387 2123556176  975149182  657774990
 [7] 1688886839 1722597766  545710096  685987118  654417054 1203404683
[13] 1011189918 2070570017 1394746157

> # or obtain the actual instances.
> iraceResults$scenario$instances[instanceID]

 [1] "./instances/2000-918.tsp" "./instances/2000-919.tsp"
 [3] "./instances/2000-519.tsp" "./instances/2000-813.tsp"
 [5] "./instances/2000-811.tsp" "./instances/2000-512.tsp"
 [7] "./instances/2000-715.tsp" "./instances/2000-819.tsp"
 [9] "./instances/2000-712.tsp" "./instances/2000-816.tsp"
[11] "./instances/2000-814.tsp" "./instances/2000-620.tsp"
```

```
[13] "./instances/2000-812.tsp" "./instances/2000-717.tsp"
[15] "./instances/2000-611.tsp"

> # If the instances are of atomic type (integers, floating-point numbers or
> # character strings), the above is similar to:
> get_instanceID_seed_pairs(iraceResults, index = pair_index, instances=TRUE)

    instanceID       seed                 instance
         <int>      <int>                   <char>
 1:         48 1498426593 ./instances/2000-918.tsp
 2:         49 1324006684 ./instances/2000-919.tsp
 3:          9  156117387 ./instances/2000-519.tsp
 4:         33 2123556176 ./instances/2000-813.tsp
 5:         31  975149182 ./instances/2000-811.tsp
 6:          2  657774990 ./instances/2000-512.tsp
 7:         25 1688886839 ./instances/2000-715.tsp
 8:         39 1722597766 ./instances/2000-819.tsp
 9:         22  545710096 ./instances/2000-712.tsp
10:         36  685987118 ./instances/2000-816.tsp
11:         34  654417054 ./instances/2000-814.tsp
12:         20 1203404683 ./instances/2000-620.tsp
13:         32 1011189918 ./instances/2000-812.tsp
14:         27 2070570017 ./instances/2000-717.tsp
15:         11 1394746157 ./instances/2000-611.tsp
```

- `experimentLog`: A matrix with columns `iteration,instance,configuration`. This matrix contains the log of all the experiments that **irace** performs during its execution. The `instance` column refers to the index of the data frame returned by `get_instanceID_seed_pairs()`. When `capping` is enabled a column `bound` is added to log the execution bound applied for each execution.

- `softRestart`: A logical vector that indicates if a soft restart was performed on each iteration. If `FALSE`, then no soft restart was performed. See option `softRestart` in Section 11.

- `state`: A list that contains the state of **irace**, the recovery (Section 8) is done using the information contained in this object. The probabilistic model of the last elite configurations can be found here by doing:

```
> # As an example, we get the model probabilities for the
> # localsearch parameter.
> iraceResults$state$model["localsearch"]

NULL

> # The order of the probabilities corresponds to:
> iraceResults$scenario$parameters$domains$localsearch

[1] "0" "1" "2" "3"
```

The example shows a list that has one element per elite configuration (ID as element name). In this case, `localsearch` is a categorical parameter and it has a probability for each of its values.

- `testing`: A list that contains the testing results. The list contains the following elements:
  - `experiments`: Matrix of experiments in the same format as the `iraceResults$experiments` matrix. The column names indicate the candidate configuration identifier and the row names contain the name of the instances.

```
> # Get the results of the testing
> iraceResults$testing$experiments

          31        9       49       72       98       96      103
1t  32749905 33073160 32930475 32986422 32761332 33004115 32934943
2t  32710952 32771521 32599919 32799886 32538536 32733625 32690959
3t  33131706 33311738 33263026 33103479 33025023 33168522 33161798
4t  32773687 32917625 32815804 32816199 32735053 32933343 32813501
5t  32805937 32954742 32959792 32885685 32649573 32963272 32834836
6t  32701639 32880447 32821284 32816098 32675167 32623539 32700062
7t  32913855 33075387 33093041 33016327 32962011 32997899 32870606
8t  32951942 33081714 33008092 33080891 32868985 32834125 32875231
9t  33097781 33264852 33036753 33091283 32837161 32927979 33013875
10t 32776387 32854968 32882947 32732776 32596401 32679623 32619241
11t 32840171 33062073 32925106 32903359 32920926 33037613 32907652
12t 32876549 32731804 32658219 32686426 32662045 32659880 32584460
13t 32792890 32784999 32766879 32770470 32687410 32736264 32735189
14t 32588302 32781490 32662935 32853708 32595427 32538913 32557994
15t 32829263 33097449 32999037 32938286 32770455 32882840 32845447
16t 33055355 33178703 33100502 33126217 32967779 32947686 33092793
17t 33395788 33397398 33246598 33236109 33116137 33117901 33166439
18t 32591674 32719003 32661651 32614252 32521541 32575740 32600864
19t 33188590 33154900 33028742 33021588 32942109 32954921 32791771
20t 32860388 32929898 33012570 32848300 32803740 32828626 32800048
21t 32696377 32948212 32836751 32717547 32669401 32662678 32757776
22t 32862244 32975313 32944747 32891111 32734522 32790878 32890641
23t 32841656 32792532 32840829 32952300 32821930 32843902 32807612
24t 32805117 32930826 32981580 32841989 32941939 32846746 32928553
25t 33163004 33149910 33165425 33126861 33004022 33039027 32996001
26t 33037596 33346731 33153912 33137718 33023771 33207244 33111313
27t 32887572 33076667 33097669 32959355 32835684 32864644 32776916
28t 32824425 33108084 32938348 32822983 32752315 32939268 32885295
29t 33144421 33244211 33295222 33244572 33235539 33279036 33117518
30t 32853822 33001860 33064180 32963126 32797929 32848020 32924648
31t 33102391 33168835 33337690 33301978 33163554 33076899 33114088
32t 32989659 33171487 33180840 32946542 33002516 33015941 32942978
33t 32993965 32921454 33002228 32936221 32689530 32936992 32998308
34t 32740503 32889667 32852592 32826153 32644257 32701105 32761568
35t 32421034 32637363 32498338 32518357 32385227 32427681 32432778
36t 32622246 32745813 32695336 32668345 32598332 32497909 32574275
```

```
37t 32552385 32740722 32558460 32526345 32410641 32486287 32475945
38t 32860528 32974748 32953906 32768454 32744922 32843640 32884432
39t 32828816 33017788 32888000 32892119 32810122 32942032 32992748
40t 32467350 32468116 32448362 32503352 32334837 32522252 32433362
41t 33228080 33209063 33295168 33055643 33036918 33040606 33105703
42t 33082756 33054510 32991354 32977229 32929594 32977750 32955021
43t 32602545 32562848 32569070 32567204 32582128 32579722 32502390
44t 32780766 32887235 32773656 32806115 32749731 32737744 32682232
45t 33009501 33124459 32949972 32911677 32755277 32829691 32804705
46t 32659139 32927255 33034700 32826872 32715119 32698481 32758583
47t 32608686 32846620 32721676 32677097 32596199 32513701 32619768
48t 32588985 32710107 32598849 32724784 32497348 32759009 32563546
49t 33237686 33300972 33162769 33220226 33077737 33201486 33229654
50t 32480717 32683174 32557868 32471396 32479003 32487112 32580473
        120      136      150      134      166      167
1t  32905174 32896944 32811340 32823802 32851188 32845151
2t  32605925 32614704 32540269 32661429 32622824 32540694
3t  33046905 32993736 32927563 33080150 33036782 33029892
4t  32846394 32833798 32805021 32784169 32748126 32723614
5t  32827010 32770313 32775025 32814691 32682028 32783382
6t  32710163 32559225 32569059 32624450 32587902 32617953
7t  32980816 32913250 32938664 32936448 32898217 32850756
8t  32877410 32896266 33021560 33054084 32923771 32835412
9t  32928827 32944697 32972745 32901321 32822132 32852044
10t 32768911 32791527 32939443 32836283 32734442 32558736
11t 32836692 32923413 32891500 32932313 33078269 32953144
12t 32662757 32573403 32584310 32573084 32613926 32586139
13t 32634940 32619179 32673727 32642861 32719418 32732651
14t 32590653 32672775 32639558 32714025 32659158 32647101
15t 32729091 32721627 32693605 32772830 32667912 32649724
16t 32867685 32985297 32930280 32954472 32983107 32798223
17t 33224345 33102141 33056812 33115169 33211388 33173750
18t 32512906 32522124 32631070 32470614 32492022 32520419
19t 33011032 33097580 32961340 32991998 33013457 33015111
20t 32726014 32725522 32604599 32689489 32732674 32801588
21t 32669751 32597092 32662213 32707333 32584059 32660232
22t 32762185 32730402 32789847 32835451 32768150 32801554
23t 32700161 32797304 32799787 32751023 32882744 32862919
24t 33002336 32917024 32878016 32848865 32833006 32850426
25t 32910777 33002641 32914969 32936249 32959208 32941455
26t 33026307 32966551 33036641 33072225 32988382 33002093
27t 32862877 32861951 32786080 32785412 32854044 32709217
28t 32709922 32913755 32687459 32843580 32782412 32799271
29t 32980873 33279886 33222418 33119808 33229746 33055143
30t 32749206 32826876 32767583 32848000 32752049 32778647
31t 33035085 33018801 33098827 33046268 33045549 33150835
32t 32889407 32893207 32904007 32861470 32929367 32801389
33t 32811658 32834105 32828819 32859057 32894892 32694159
```

```
34t 32661073 32734482 32650003 32605003 32653877 32637915
35t 32383338 32442487 32431843 32472772 32420295 32483555
36t 32471891 32570809 32649049 32534293 32425453 32601710
37t 32321347 32387127 32465270 32532289 32444685 32414100
38t 32712182 32667527 32679216 32767449 32580279 32710671
39t 32779845 32777496 32823209 32883748 32758570 32766333
40t 32370800 32374213 32277047 32423130 32432412 32349933
41t 32989404 33073463 32902621 32973099 33086439 32978788
42t 32922033 32881460 32978073 32882100 33035102 32945336
43t 32474660 32485402 32661308 32561604 32497556 32476916
44t 32677053 32680745 32630216 32785578 32528169 32731436
45t 32750872 32771565 32741031 32820493 32863595 32741179
46t 32583598 32727063 32694971 32703910 32712218 32685928
47t 32465397 32551850 32497368 32588926 32540344 32545684
48t 32555548 32540113 32517745 32635332 32512177 32504362
49t 33179838 33121790 33087736 33078508 33120681 33048639
50t 32423664 32433328 32440086 32345774 32485926 32410608
```

– `seeds`: The seeds used for the experiments, each seed corresponds to each instance in the rows of the test `experiments` matrix.

```
> # Get the seeds used for testing
> iraceResults$testing$seeds

        1t         2t         3t         4t         5t         6t
1987246585  415084885  310098509 1191891037 1992920706 1556635314
        7t         8t         9t        10t        11t        12t
 660937021  627123935  421925618 1440918674  415630090  786655150
       13t        14t        15t        16t        17t        18t
 984347817  211089707 2043267046  911388040 1946490076 1569928577
       19t        20t        21t        22t        23t        24t
1187926914  467003972 1143235327 2139328848 1336400162  197579087
       25t        26t        27t        28t        29t        30t
1622161044 1015660437 1916222549 1933809701 1780651132  527334133
       31t        32t        33t        34t        35t        36t
 879873640 2067737294 1608280244  393822372  932726103 1590808222
       37t        38t        39t        40t        41t        42t
1982372176  327392830 1143693767  120344916 1498426593 1324006684
       43t        44t        45t        46t        47t        48t
 156117387 2123556176  975149182  657774990 1688886839 1722597766
       49t        50t
 545710096  685987118
```

In the example, instance `1000-1.tsp` is executed with seed 1987246585.

9.3 Analysis of results

The final configurations returned by **irace** are the elites of the final race. They are reported in decreasing order of performance, that is, the best configuration is reported first.

If testing is performed, you can further analyze the resulting best configurations by performing statistical tests in R:

```
> results <- iraceResults$testing$experiments
> # Wilcoxon paired test
> conf <- gl(ncol(results), # number of configurations
+            nrow(results), # number of instances
+            labels = colnames(results))
> pairwise.wilcox.test (as.vector(results), conf, paired = TRUE, p.adj = "bonf")

 Pairwise comparisons using Wilcoxon signed rank test with continuity correction

data:  as.vector(results) and conf

    31      9       49      72      98      96      103     120
9   7.3e-06 -       -       -       -       -       -       -
49  0.01766 0.01576 -       -       -       -       -       -
72  1.00000 4.7e-05 0.71373 -       -       -       -       -
98  0.00027 1.3e-07 7.7e-08 7.6e-07 -       -       -       -
96  1.00000 9.1e-07 0.00237 0.22987 0.00781 -       -       -
103 1.00000 1.2e-07 9.0e-06 0.00666 0.05112 1.00000 -       -
120 5.2e-05 7.3e-08 1.9e-07 7.2e-07 1.00000 0.00058 0.00502 -
136 0.00281 7.7e-08 2.4e-07 1.0e-06 1.00000 0.01459 0.03841 1.00000
150 0.00237 1.9e-07 3.8e-07 4.1e-06 1.00000 0.01071 0.05486 1.00000
134 0.03446 6.1e-08 1.3e-06 1.3e-06 1.00000 0.04434 0.31347 1.00000
166 0.00018 1.1e-07 8.1e-07 2.5e-06 1.00000 0.01251 0.11268 1.00000
167 2.6e-05 7.3e-08 1.0e-07 1.1e-07 1.00000 0.00029 0.00024 1.00000
    136     150     134     166
9   -       -       -       -
49  -       -       -       -
72  -       -       -       -
98  -       -       -       -
96  -       -       -       -
103 -       -       -       -
120 -       -       -       -
136 -       -       -       -
150 1.00000 -       -       -
134 1.00000 1.00000 -       -
166 1.00000 1.00000 1.00000 -
167 1.00000 1.00000 0.33319 1.00000

P value adjustment method: bonferroni
```

The Kendall concordance coefficient (`W`) and the Spearman's rho can be applied over data that has the characteristics of the data obtained in the testing, that is a full matrix where all configurations are executed in all instances. `W` can show if the configurations tested have an homogeneous performance on the used instances set. If evidence of an heterogeneous scenario found we recommend to make some adjustments in the **irace** options as described in Section 10.6.

```
> irace:::concordance(iraceResults$testing$experiments)

$kendall.w
[1] 0.4517495

$spearman.rho
[1] 0.4405607
```

It is also possible, as shown in Fig. 7, to plot the performance on the test set of the best-so-far configuration over the number of experiments as follows:

```
# Get summary data from the logfile.
irs <- irace_summarise(iraceResults)
# Get number of iterations
iters <- irs$n_iterations
# Get number of experiments (runs of target-runner) up to each iteration
fes <- cumsum(table(iraceResults$state$experiment_log[["iteration"]]))
# Get the mean value of all experiments executed up to each iteration
# for the best configuration of that iteration.
elites <- as.character(iraceResults$iterationElites)
values <- colMeans(iraceResults$testing$experiments[, elites])
stderr <- function(x) sqrt(var(x)/length(x))
err <- apply(iraceResults$testing$experiments[, elites], 2L, stderr)
plot(fes, values, type = "s",
     xlab = "Number of runs of the target algorithm",
     ylab = "Mean value over testing set", ylim=c(min(values-err),max(values+err)))
points(fes, values, pch=19)
arrows(fes, values - err, fes, values + err, length=0.05, angle=90, code=3)
text(fes, values, elites, pos = 1)
```

The **irace** package also provides an implementation of the ablation method [6]. See Section 10.10.

Finally, more advanced visualizations of the behavior of **irace** are provided by the ACVIZ software package [4], which is available at `https://github.com/souzamarcelo/acviz`. See an example in Fig. 8.

10 Advanced topics

10.1 Tuning budget

Before setting the budget for a run of **irace**, please consider the number of parameters that need to be tuned, available processing power and available time. The optimal budget depends on the difficulty of the tuning scenario, the size of the parameter space and the heterogeneity of the instances. Typical values range from 1 000 to 100 000 runs of the target algorithm, although smaller and larger values are also possible. Currently, **irace** does not detect whether the given budget allows generating all possible configurations. In such a case, the use of *iterated* racing is unnecessary: One can simply perform a single race of all configurations (see FAQ in Section 12.11).

Irace provides two options for setting the total tuning budget (`maxExperiments` and `maxTime`). The option `maxExperiments` limits the number of executions of `targetRunner` performed by

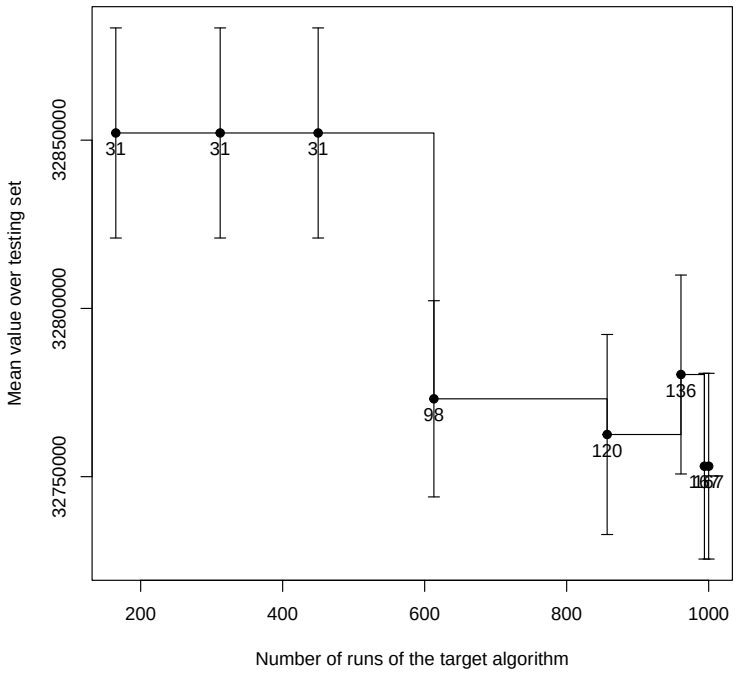


Figure 7: Testing set performance of the best-so-far configuration over number of experiments. Label of each point is the configuration ID.

irace. The option `maxTime` limits the total time of the `targetRunner` executions. When this latter option is used, `targetRunner` must return the evaluation cost together with the execution time (`"cost time"`).



When the goal is to minimize the computation time of an algorithm, and you wish to use `maxTime` as the tuning budget, `targetRunner` must return the time also as the evaluation cost, that is, return the time twice as `"time time"`.



When using `targetEvaluator` and using `maxTime` as tuning budget, `targetRunner` just returns the time (`"time"`) and `targetEvaluator` returns the cost.

When using `maxTime`, **irace** estimates the execution time of each `targetRunner` execution before the configuration. The amount of budget used for the estimation is set with the option `budgetEstimation` (default is 2%). The obtained estimation is adjusted after each iteration using the obtained results and it is used to estimate the number of experiments that can be executed. Internally, **irace** uses the number of remaining experiments to adjust the number of configurations tested in each race.

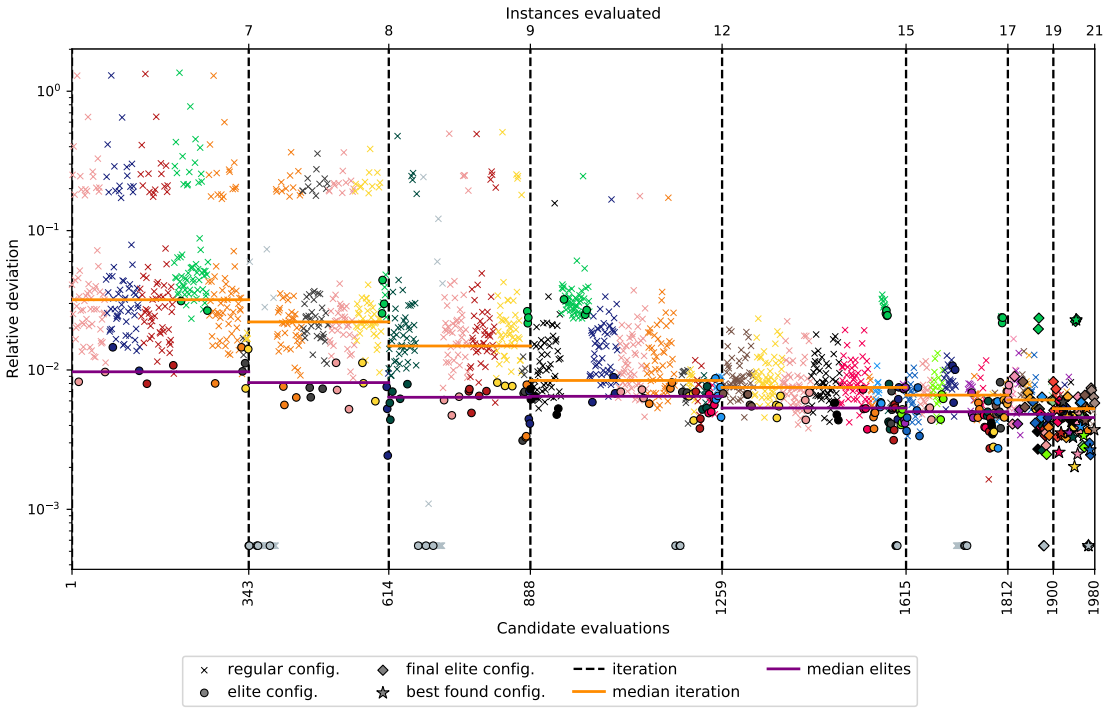


Figure 8: Visualization produced by ACVIZ [4].

10.2 Multi-objective tuning

Currently, **irace** only optimizes one cost value at a time, which can be solution cost, computation time or any other objective that is returned to **irace** by the `targetRunner`. If the target algorithm is multi-objective, it will typically return not a single cost value, but a set of objective vectors (typically, a Pareto front). For tuning such a target algorithm with **irace**, there are two alternatives. If the algorithm returns a single vector of objective values, they can be aggregated into one single number by using, for example, a weighted sum. Otherwise, if the target algorithm returns a set of objective vectors, a unary quality metric (e.g., the hypervolume) may be used to evaluate the quality of the set.[3]

The use of aggregation or quality metrics often requires normalizing the different objectives. If normalization bounds are known a priori for each instance, normalized values can be computed by `targetRunner`. Otherwise, the bounds may be dynamically computed while running **irace**, by using `targetEvaluator`. In this case, `targetRunner` will save the output of the algorithm, then the first call to `targetEvaluator` will examine the output produced by all calls to `targetRunner` for the same instance, update the normalization bounds and return the normalized output. Subsequent calls to `targetEvaluator` for the same instance will simply return the normalized output. A similar approach can be used to dynamically compute the reference points or sets often required by unary quality metrics.

For more information about defining a `targetEvaluator`, see Section 5.3. Examples of tuning a multi-objective target algorithm using the hypervolume can be found in the examples at

[3]Multiple metrics are implemented in C, R and Python in the `moocore` package (https://github.com/multi-objective/moocore).

`$IRACE_HOME`/examples/hypervolume and `$IRACE_HOME`/examples/moaco.

10.3 Improving anytime behavior

In some scenarios, we may be unsure when the end-user may terminate the target-algorithm, and we would like to tune the parameters of the algorithm so that it finds a solution as good as possible for any given termination criterion. That is, we would like to obtain a parameter configurations with good *anytime performance*.

Following the approach proposed by López-Ibáñez and Stützle [10], **irace** can tune an algorithm to improve its anytime performance. For that purpose, the algorithm must output a (time, solution cost) pair every time that the best-so-far solution within a run is improved. Such pairs form a matrix with two columns and as many rows as improvements were made. We can define a *reference point*, where the first component is value slightly higher than the maximum runtime of a single run and the second component is a maximum cost value such that differences above that maximum are irrelevant for evaluating performance. Using this reference point and the matrix as an input, one can calculate the hypervolume of the matrix, as described above in Section 10.2. The negative of this hypervolume value becomes the cost returned by the `targetRunner` to **irace**.

It is also possible to specify preferences over the trade-off between runtime and solution cost by using the weighted hypervolume. Further details and examples can be found in López-Ibáñez and Stützle [10].

Although there are other ways to measure anytime performance, most of them are equivalent or an approximation of the measure described above [14].

10.4 Tuning for minimizing computation time

When using **irace** for tuning algorithms that only report computation time to reach a target, `targetRunner` should return the execution time of a configuration instead of solution cost. When using `maxTime` as the budget, this means that `targetRunner` must return twice the execution time since the first value is the minimization objective and the second value is used to track the budget consumed.

Starting from version 3.0, **irace** includes an elitist racing procedure that implements an **adaptive capping mechanism** [16]. Adaptive capping [7] is a configuration technique that avoids the execution of long runs of the target algorithm, focusing the configuration budget in the evaluation of the best configurations found. This is done by bounding the execution time of each configuration based on the best performing candidate configurations.

To use adaptive capping, the `capping` option must be enabled and the `elitist` irace option must be selected. When evaluating candidate configurations on an instance, **irace** calculates an execution bound based on the execution times of the elite configurations. The `boundType` option defines how the performance of the elite configurations is defined to obtain the execution bound. The default value of `boundType` calculates the performance (p_i^s) of each elite configuration (s) as the mean execution time of the instances already executed in the race and the currently executed instance (i). The `cappingType` option specifies the measure used to obtain the elite configurations bound. By default, the execution bound is calculated as the median of the execution times of the elite configurations:

$$b_i = \text{Median}_{\theta_s \in \Theta^{\text{elite}}}\{p_i^s\} \tag{1}$$

The execution bound for new configurations (j) is calculated by multiplying the elite configurations bound by the number of instances (i) in the execution list and subtracting the mean execution time of the instances executed by the candidate:

$$k_i^{'j} = b_i \cdot i + b^{\min} - p_{i-1}^j \cdot (i-1) \tag{2}$$

A small constant $b^{\min}$ is added to account for time measurements errors. These settings are also used to apply a dominance elimination criterion together with the statistical test elimination. The domination criterion is defined as:

$$b_i + b^{\min} < p_i^j \tag{3}$$

When elite configurations dominate new configurations, these are eliminated from the race.



The default statistical test when `capping` is enabled is `t-test`. This test is more appropriate to configure algorithms for optimizing runtime (see Section 10.7).

The execution bound is constantly adjusted by **irace** based on the best configurations times, nevertheless, a maximum execution time ($b^{\max}$) is never exceeded. This maximum execution time must be defined in the configuration scenario when `capping` is enabled. To specify the maximum execution bound for the target runner executions use the `boundMax` option. The final execution bound (k_i^j) is calculated by:

$$k_i^j = \begin{cases} b^{\max} & \text{if } {k_i'}^j > b^{\max}, \\ \min\{b_i, b^{\max}\} & \text{if } {k_i'}^j \leq 0, \\ {k_i'}^j & \text{otherwise;} \end{cases} \tag{4}$$

Additionally, the `boundDigits` option defines the precision of the time bound provided by **irace**, the default setting is 0.

Timed out executions occur when the maximum execution bound (`boundMax`) is reached and the algorithm has not achieved successful termination or a defined quality goal. In this case, it is a common practice to apply a penalty known as PARX, in which timeouts are penalized by multiplying `boundMax` by a constant X. The constant X may be set using the `boundPar` option. Bounded executions are executions that do not achieve successful termination or a defined quality goal in the execution bound (k_i^j) set by **irace**, which is smaller than `boundMax`. The `boundAsTimeout` option replaces the evaluation of bounded executions by the `boundMax` value. More details about the implementation of adaptive capping can be found in Pérez Cáceres et al. [16].



Note that bounded executions are not timed out executions and thus, they will not be penalized by PARX.



Penalized evaluations of timed out and bounded executions are only used for the elimination tests and the comparison between the quality of configurations. To calculate execution bounds and computation budget consumed, **irace** uses only unpenalized execution times. The unpenalized execution time must be provided by the target runner or target evaluator as described in Section 5.2 and Section 5.3 .



More advanced capping methods that are applicable to minimizing solution cost are available when combining irace with the `capopt` package described by De Souza et al. [5].

10.5 Hyper-parameter optimization of machine learning methods

The **irace** package can also be used for model selection and hyper-parameter optimization of machine learning (ML) methods. We will next explain a possible setup for one given dataset and

using 10-fold cross-validation (CV). Generalizing to multiple datasets and different resampling strategies, e.g. leave-one-out, is straightforward.

First, split the dataset into training, to be used by **irace**, and testing, to be used for evaluating the performance of the configuration returned by **irace**. A typical split could be 70% and 30%, respectively.

The training set is used by **irace** to perform 10-fold CV, that is, the data is split into 10 folds. A single run of the `targetRunner` will use 9 folds for training and the remaining fold for validation. Splitting the data into folds can be done at each call of `targetRunner` or before running **irace**, however, it is important that the split is always the same for every call of the `targetRunner`, i.e., the content of the folds does not change, only which folds are used for training and validation will change.

The setup of **irace** should be as follows:

- `trainInstancesFile="train-instances.txt"`, where this file contains one number per line from 1 to 10. This number will tell the `targetRunner` which fold should be used for validation.
- `trainInstancesDir=""`, because the folds are the "instances" and you do not have actual instance files. If you want to pass the name of the dataset to the `targetRunner`, you can specify it either at each line of `"train-instances.txt"`, directly in the `targetRunner`, or as a fixed parameter in the `parameterFile`.
- `deterministic=1` unless it really makes sense to train more than once the same ML model on the same data. If it makes sense, then your `targetRunner` should use the seed passed by **irace** to seed the ML model before training.
- `sampleInstances=0` because the folds should already be generated by randomly sampling the dataset.
- `testType="t-test` because the performance metrics in ML are typically the mean of the CV results, which assumes that the performance are close to normally distributed.
- `firstTest=2` because **irace** should discard configurations very aggressively looking for maximum generality.

Finally, your `targetRunner` needs to be able to do the following:

- Receive from **irace** the hyper-parameter settings, the dataset name and a fold number (the "instance"). Let us use fold 3 as an example.
- Train the ML model on the whole training set minus fold 3, then validate (score) the model on fold 3 and return the score to **irace** (negated if the score must be maximized, because **irace** assumes minimization). Since each fold is different, each instance should give a different result. Each row in the table printed by **irace** should print something different; otherwise, something is wrong in your setup.

The above is actually 10 times faster than doing 10-fold CV for each call to `targetRunner`, thus, you should assign to **irace** 10 times the budget than what would be assigned to other methods that do a complete 10-fold CV at each step.

10.6 Heterogeneous scenarios

We classify a scenario as homogeneous when the target algorithm has a consistent performance regarding the instances; roughly speaking, good configurations tend to perform well and bad configurations tend to perform poorly on all instances of the problem. By contrast, in heterogeneous

scenarios, the target algorithm has an inconsistent performance on different instances, that is, some configurations perform well for a subset of the instances, while they perform poorly for a different subset.

When facing a heterogeneous scenario, the first question should be whether the objective of tuning is to find configurations that perform reasonably well over all instances, even if that configuration is not the best ones in any particular instance (a generalist). If this is not the goal, then it would be better to partition instances into more similar subsets and execute **irace** separately on each subset. This will lead to a portfolio of algorithm configurations, one for each subset, and algorithm selection techniques can be used to select the best configuration from the portfolio when facing a new instance.

To make sure **irace** is not misled by results on few instances, it may be useful to increase the number of instances executed before doing a statistical test using the option `firstTest`, e.g., `--first-test 10` (default value is 5), in order to see more instances before discarding configurations. The option `elitistNewInstances` in elitist **irace** (option `elitist`) can be used to increase the number of new instances executed in each iteration, e.g., `--elitist-new-instances 5` (default value is 1).

If finding an overall good configuration for all the instances is the objective, then we recommend that instances are randomly sampled (option `sampleInstances`), unless one can provide the instances in a particular order that does not bias the tuning towards any subset.

If instances are easily categorized in different classes, then we recommend to create "blocks" of instances in `trainInstancesFile`, where each block should contain one instance from each class. Then set the option `blockSize` to the number of classes within each block, so that **irace** will always see a complete block of instances before eliminating configurations. The value of `blockSize` will multiply the values of `firstTest`, `eachTest` and `elitistNewInstances`. Randomly sampling instances (`sampleInstances=1`) will randomly sample the blocks but not break the blocks.

While executing **irace**, the homogeneity of the scenario can be observed by examining the values of Spearman's rank correlation coefficient and Kendall's concordance coefficient in the text output of **irace**. See Section 9.1 for more information.

10.7 Choosing the statistical test

The statistical test used in **irace** identifies statistically bad performing configurations that can be discarded from the race in order to save budget. Different statistical tests use different criteria to compare the cost of the configurations, which has an effect on the tuning results.

Irace provides two types of statistical tests (option `testType`). Each test has different characteristics that are beneficial for different goals:

- Friedman test (`F-test`): This test uses the ranking of the configurations to analyze the differences between their performance. This makes the test suitable for scenarios where the scale of the performance metric is not as important to assess configurations as their relative ranking. This test is also indicated when the distribution of the mean performances deviates greatly from a normal distribution. For example, the ranges of the performance metric on different instances may be completely difference and comparing the performance of different configurations using the mean over multiple instances may be deceiving. We recommend to use the `F-test` (default when `capping` is not enabled) when tuning for solution cost and whenever the best performing algorithm should be among the best in as many instances as possible.

- Student's t-test (`t-test`): This test uses the mean performance of the configurations to

analyze the differences between the configurations.[4] This makes the test suitable for scenarios where the differences between values obtained for different instances are relevant to assess good configurations. We recommend using t-test, in particular, when the target algorithm is minimizing computation time and, in general, whenever the best configurations should obtain the best average solution cost.

The confidence level of the tests may be adjusted by using the option `confidence`. Increasing the value of `confidence` leads to a more strict statistical test. Keep in mind that a stricter test will require more budget to identify which configurations perform worse. A less strict test discards configurations faster by requiring less evidence against them and, therefore, it is more likely to discard good configurations.

10.8 Complex parameter space constraints

Some parameters may have complex dependencies. Ideally, parameters should be defined in the way that is more likely to help the search performed by **irace**. For example, when tuning a branch and bound algorithm, one may have the following parameters:

- branching (`b`) that takes values in `{0,1,2,3}`, where 0 indicates no branching will be used and the rest are different types of branching.
- stabilization (`s`) that takes values in `{0,1,2,3,4,5,6,7,8,9,10}`, of which for `b=0` only `{0,1,2,3,4,5}` are relevant.

In this case, it is not possible to describe the parameter space by defining only two parameters for **irace**. An extra parameter must be introduced as follows:

```
# name    label   type   range                     condition
b         "-b "   c      (0,1,2,3)
s1        "-s "   c      (0,1,2,3,4,5)             | b == "0"
s2        "-s "   c      (0,1,2,3,4,5,6,7,8,9,10)  | b != "0"
```

Parameters whose values depend on the value of other parameters may also require using extra parameters or changing the parameters and processing them in `targetRunner`. For example, given the following parameters:

- Population size (`p`) takes the integer values $[1, 100]$.
- Selection size (`s`) takes the same values but no more than the population size, that is [1,`p`].

In this case, it is possible to describe the parameters `p` and `s` using surrogate parameters for **irace** that represent a ratio of the original interval as follows:

```
# name    label   type   range
p         "-p "   i      (1,100)
s_f       "-s "   r      (0.0,1.0)
```

and `targetRunner` must calculate the actual value of `s` as $\min(\max(\text{round}(\texttt{s_f} \cdot \texttt{p}, 1)), 100)$. For example, if the parameter `p` has value 50 and the surrogate parameter `s_f` has value 0.3, then `s` will have value 15.

The processing within `targetRunner` can also split and join parameters. For example, assume the following parameters:

[4] The t-test does not require that the performance values follow a normal distribution, only that the distribution of sample means does. In practice, the t-test is robust despite large deviations from the assumptions.

```
# name    label   type   range
m         "-m "   i      (1,250)
e         "-e "   r      (0.0,2.0)
```

These parameters could be used to define a value $\texttt{m} \cdot 10^{\texttt{e}}$ for another parameter (`--strength`) not known by **irace**. Then, `targetRunner` takes care of parsing `-m` and `-e`, computing the strength value and passing the parameter `--strength` together with its value to the target algorithm.

More complex parameter space constraints may be implemented by means of the `repairConfiguration` function (Section 5.6).

10.9 Unreliable target algorithms and immediate rejection

There are some situations in which the target algorithm may fail to execute correctly. By default, **irace** stops as soon as a call to `targetRunner` or `targetEvaluator` fails, which helps to detect bugs in the target algorithm. Sometimes the failure cannot be fixed because it is due to system problems, network issues, memory limits, bugs for which no fix is available, or fixing them is impossible because there is no access to the source code.

In those cases, if the failure is caused by random errors or transient system problems, one may wish to ignore the error and try again the same call in the hope that it succeeds. The option `targetRunnerRetries` indicates the number of times a `targetRunner` execution is repeated if it fails. Use this option only if you know additional repetitions could be successful.

If the target algorithm consistently fails for a particular set of configurations, these configurations may be declared as forbidden (Section 5.1.5) so that **irace** avoids them. On the other hand, if the configurations that cause the problem are unknown, the `targetRunner` should return `Inf` so that **irace** immediately rejects the failing configuration. This immediate rejection should be used with care according to the goals of the tuning. For example, a configuration that crashes on a particular instance, e.g., by running out of memory, might still be considered acceptable if it gives very good results on other instances. The configurations which were rejected during the execution of **irace** are saved in the Rdata output file (see Section 9.2).



If the configuration budget is specified in total execution time (`maxTime` option), immediate rejected executions must provide the cost and time (which must be `Inf 0`). Nevertheless, rejected configurations will be excluded from the execution time estimation and the execution bound calculation.

10.10 Ablation Analysis

The ablation method [6] takes two configurations (source and target) and generates a sequence of configurations that differ between each other just in one parameter, where parameter values in source are replaced by values from target. The sequence can be seen as a "path" from the source to the target configuration. This can be used to find new better "intermediate" configurations or to analyse the impact of the parameters in the performance.

To perform ablation, you can use the `ablation()` R function

```
ablog <- ablation("irace.Rdata", src = 1)
plotAblation(ablog)
```

or the `ablation` command-line executable (see more details below).

```
ablation -l irace.Rdata --src 1 --plot plot-ablation.pdf --plot-type rank,boxplot
```

You may specify the IDs of the source and target configurations. By default, the source is taken as the first configuration evaluated by **irace** and the target as the best overall configuration found. Use the function `plotAblation` to visualize the ablation results (Fig. 9).

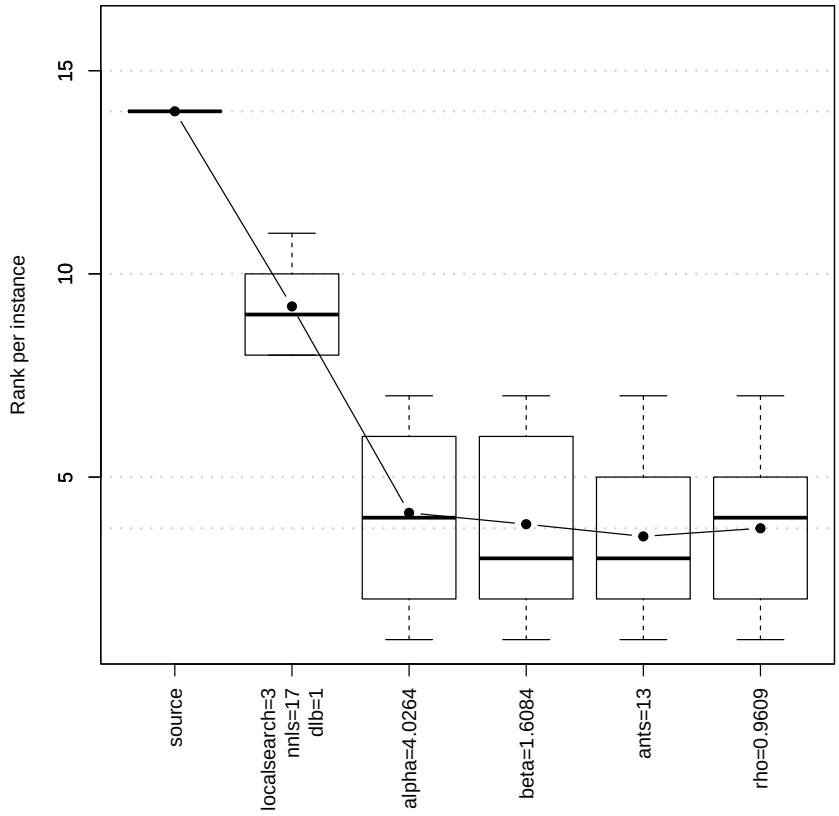


Figure 9: Example of plot generated by `plotAblation()`.

The function returns a list containing the following elements:

`configurations`: A dataframe of configurations tested during ablation.

`instances`: The instances used for the ablation.

`scenario`: Scenario options provided by the user.

`trajectory`: Best configuration IDs at each step of the ablation.

`best`: Best overall configuration found.

We also provide a command-line executable (`ablation.exe` in Windows) that allows you to perform ablation without launching R. It is installed in the same location as the **irace** command-line executable and has the following options:

```
#------------------------------------------------------------------------------
# ablation: An implementation in R of Ablation Analysis
```

```
# Version: unknown
# Copyright (C) 2020--2025
# Manuel Lopez-Ibanez     <manuel.lopez-ibanez@manchester.ac.uk>
# Leslie Perez Caceres    <leslie.perez.caceres@ulb.ac.be>
#
# This is free software, and you are welcome to redistribute it under certain
# conditions.  See the GNU General Public License for details. There is NO
# WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
#-------------------------------------------------------------------------------
# installed at: /tmp/RtmpsdfIeQ/Rinst1b7b476d23b9/irace
# called with: --help
-l,--log-file            Path to the (.Rdata) file created by irace from which
                         the "iraceResults" object will be loaded.
-S,--src                 Source configuration ID or the path to a file
                         containing the configuration. Default: 1.
-T,--target              Target configuration ID (by default the best
                         configuration found by irace) or the path to a file
                         containing the configuration.
-P,--params              Specific parameter names to be used for the ablation
                         (separated with commas). By default use all
-t,--type                Type of ablation to perform: "full" will execute each
                         configuration on all "--n-instances" to determine the
                         best-performing one; "racing" will apply racing to
                         find the best configurations. Default: full.
-n,--nrep                Number of replications per instance used in "full"
                         ablation. Default: 1.
   --seed                Integer value to use as seed for the random number
                         generation. Default: 1234567.
-o,--output-file         Log file to save the ablation log. If "", the results
                         are not saved to a file. Default: log-ablation.Rdata.
   --instances-file      Instances file used for ablation: "train", "test" or a
                         filename containing the list of instances. Default:
                         train.
-p,--plot                Output filename (.pdf) for the plot. If not given, no
                         plot is created.
-O,--plot-type           Type of plot. Supported values are "mean", "boxplot",
                         "rank" or "rank,boxplot". Default: mean.
   --old-path            Old path found in the log-file (.Rdata) given as input
                         to be replaced by --new-path.
   --new-path            New path to replace the path found in the log-file
                         (.Rdata) given as input.
-e,--exec-dir            Directory where the target runner will be run.
-s,--scenario            Scenario file to override the scenario given in the
                         log-file (.Rdata)
   --parallel            Number of calls to targetRunner to execute in
                         parallel. Values 0 or 1 mean no parallelization.
```

10.11 Post-selection race

At the end of the configuration process, **irace** will automatically run a post-selection race [18] to try to consume any remaining budget. This post-selection race will not generate new configurations. It will only do additional runs on configurations already evaluated in the current run of **irace**. First, it will try to evaluate the best configurations found in new instances not seen yet. If this is not possible due to the budget available, then it will evaluate configurations that have been evaluated on fewer instances than the best configuration found. Currently, this automatic post-selection is not supported when the budget is measured in runtime (`maxTime`. The automatic post-selection can be disabled using the option `postselection`.

It is also possible to manually run a post-selection race after completing a run of **irace** using the R function `psRace()`.

```
# Execute all elite configurations in the iterations
psRace("irace.Rdata", max_experiments = 0.5, iteration_elites=TRUE)
# Execute a set of configurations IDs providing budget
psRace("irace.Rdata", conf_ids = c(34, 87, 102, 172, 293), max_experiments = 500)
```

The execution of the post-selection race adds an element (`psrace_log`) to the `iraceResults` object saved in the **irace** log file (`logFile`). The post-selection log consists of a list with the following elements:

`configurations`: Configurations used in the post-selection race.

`instances`: Data frame with the instances and seeds used in the post-selection race.

`max_experiments`: Configuration budget assigned to the post-selection race.

`experiments`: Matrix of results generated by the post-selection race, in the same format as the matrix `iraceResults$experiments`. Column names are the configuration IDs and row names are the instance IDs.

`elites`: Best configuration(s) found in the post-selection race.

10.12 Parameter importance analysis using PyImp

The **PyImp**[5] tool developed by the AutoML group[6] supports various parameter importance analysis methods using surrogate models. Given a performance dataset of an algorithm configuration scenario, a Random Forest is built to predict performance of all algorithm configurations. Parameter importance analyses are then applied on the prediction model. The model serves as a surrogate for the original target algorithm, so that the algorithm does not need to be executed during the analyses. Three analysis methods are supported, namely fANOVA [9] (functional analysis of variance), forward selection [8], and ablation analysis with surrogates [1]. Note that the **irace** package directly supports ablation (without surrogate models) analysis with and without racing (Section 10.10). Although ablation analysis without surrogates may be more time-consuming, results of the surrogate version may be less accurate than the non-surrogate one.

The `.Rdata` dataset generated by **irace** can be used as input for **PyImp**. The package `irace2pyimp`[7] is able to convert an `irace.Rdata` file into the input format required by **PyImp**. The conversion can be accessed either through the R console (function `irace2pyimp`),

[5] https://github.com/automl/ParameterImportance
[6] https://www.automl.org/
[7] https://github.com/ndangtt

or via command line executable `irace2pyimp` (or `irace2pyimp.exe` in Windows) provided by the package. You can find the location of the executable by running this command in R: `system.file(file.path("bin", "irace2pyimp"), package="irace2pyimp", mustWork=TRUE)` or `system.file(file.path("bin", "irace2pyimp.exe"), package="irace2pyimp", mustWork=TRUE)` in Windows.

To see the usage of the executable, please run: `irace2pyimp --help`. For more information on the R function `irace2pyimp`, type in the R console: `?irace2pyimp`.

Given as input an `irace.Rdata` file, the script will generate the following output files:

- `params.pcs`: a text file containing the parameter space definition.
- `runhistory.json`: a JSON file containing the list of algorithm configurations evaluated during the tuning and the performance data obtained.
- `traj_aclib2.json`: a JSON file containing the best configurations after each iteration of **irace**. The last configuration will be used as the target configuration in ablation analysis.
- `scenario.txt`: a text file containing the definition of the tuning scenario.
- `instances.txt`: a text file containing the list of instances.
- `features.csv`: a .csv file containing instance features. If no instance features are provided, the index of each instance will be used as a feature.

PyImp can then be called using the files listed above as input. Several examples on how to use the script and call **PyImp** can be found at `system.file("/examples/",package="irace2pyimp")`.

10.13 Monitoring iterations with a callback

The scenario option `iterationCallback` may be set to an R function that is called after every completed **irace** iteration. For example, the following callback reports the consumed budget and the identifier of the best configuration found in each iteration:

```
report_progress <- function(iteration, elites, progress, ...)
{
  if (progress$maxTime > 0) {
    used <- progress$timeUsed
    total <- progress$maxTime
    budget_type <- "time"
  } else {
    used <- progress$experimentsUsed
    total <- progress$maxExperiments
    budget_type <- "experiments"
  }
  message("Iteration ", iteration, "/", progress$nbIterations,
          ": ", used, "/", total, " ", budget_type,
          " used; best configuration: ", elites[[".ID."]][1L])
}
scenario$iterationCallback <- report_progress
irace(scenario)
```

The callback is called with the named arguments `iteration`, `elites`, and `progress`. User-defined callbacks should use these exact argument names and take `...` as their last argument so they remain compatible if future versions of **irace** provide more information. The `elites` data frame is ordered from best to worst and is copied before it is passed to the callback.

The `progress` list contains `nbIterations`, `maxExperiments`, `experimentsUsed`, `remainingBudget`, `remainingBudgetEstimated`, `currentBudget`, `currentBudgetUsed`, `maxTime`, `timeUsed`, `remainingTime`, and `boundEstimate`. The budgets `remainingBudget`, `currentBudget`, and `currentBudgetUsed` are numbers of target-algorithm executions. When `maxTime` is used, `remainingBudget` is estimated from `boundEstimate`, and `remainingBudgetEstimated` is `TRUE`. In that mode, `remainingTime` uses the same unit as `maxTime`; otherwise, `remainingTime` and `boundEstimate` are `NA`.

The callback return value is ignored. Errors raised by the callback stop **irace**. Random-number generation within the callback does not affect the random-number sequence of the tuning run.

11 List of command-line and scenario options

Most **irace** options can be specified in the command line using a flag or in the **irace** scenario file using the option name (or setting their value in the `scenario` list passed to the various R functions exported by the package). This section describes the various **irace** options that can be specified by the user in this way.



Relative filesystem paths (e.g., `../scenario/`) given in the command-line are relative to the current working directory (the directory at which **irace** is invoked). However, paths given in the scenario file are relative to the directory containing the scenario file. See also Table 1.

11.1 General options

`--help` *flag:* `-h` *or* `--help` *default:*
Show the list of command-line options of **irace**.

`--version` *flag:* `-v` *or* `--version` *default:*
Show the version of **irace**.

`--check` *flag:* `-c` *or* `--check` *default:*
Check that the scenario and parameter definitions are correct and test the execution of the target algorithm. See Section 4.

`--init` *flag:* `-i` *or* `--init` *default:*
Initialize the working directory with the template config files. This copies the files in `$IRACE_HOME/templates` to the working directory without overwriting the files with the same names as those of the template files.

`scenarioFile` *flag:* `-s` *or* `--scenario` *default:* `./scenario.txt`
File that contains the scenario setup and other irace options. All options listed in this section can be included in this file. See `$IRACE_HOME/templates/` for an example. Relative file-system paths specified in the scenario file are relative to the scenario file itself.

`execDir` *flag:* `--exec-dir` *default:* `./`
Directory where the target runner will be run. The default execution directory is the current directory.



The execution directory must exist before executing **irace**, it will not be created automatically.

`logFile` *flag:* `-l` *or* `--log-file` *default:* `./irace.Rdata`
File to save tuning results as an R dataset. The provided path must be either an absolute path or relative to `execDir`. See Section 9.2 for details on the format of the R dataset.

`quiet` *flag:* `-q` *or* `--quiet` *default:* `0`
Reduce the output generated by irace to a minimum.

`debugLevel` *flag:* `--debug-level` *default:* `0`
Level of information to display in the text output of **irace**. A value of 0 silences all debug messages. Higher values (1, 2 or 3) provide more verbose debug messages. Details about the text output of **irace** are given in Section 9.1.

`seed` *flag:* `--seed` *default:*
Seed to initiallize the random number generator. The seed must be a positive integer. If the seed is `""` or `NULL`, a random seed will be generated.

`repairConfiguration` *default:*
User-defined R function that takes a configuration generated by **irace** and repairs it. See Section 5.6 for details.

`iterationCallback` *default:*
Optional R function called after each **irace** iteration with the iteration number, ordered elite configurations, and progress information. See Section 10.13 for details.

`postselection` *flag:* `--postselection` *default:* `1`
Perform a postselection race after the execution of **irace** to consume all remaining budget (requires elitist racing). Value 0 disables the postselection race. See Section 10.11.

`aclib` *flag:* `--aclib` *default:* `0`
Enable/disable AClib mode. This option enables compatibility with `GenericWrapper4AC` (`https://github.com/automl/GenericWrapper4AC/`) as `targetRunner` script.

11.2 Elitist irace

`elitist` *flag:* `-e` *or* `--elitist` *default:* `1`
Enable/disable elitist **irace**.

In the **elitist** version of `irace` [12], elite configurations are not discarded from the race until non-elite configurations have been executed on the same instances as the elite configurations.

Each race begins by evaluating all configurations on a number of new instances. This number is defined by the option `elitistNewInstances`. After the new instances have been evaluated, configurations are evaluated on instances seen in the previous race. Elite configurations already have results for most of these previous instances and, therefore, do not need to be re-evaluated. Finally, after configurations have been evaluated on all these instances, the race continues by evaluating additional new instances.

The statistical tests can be performed at any moment during the race according to the setting of the options `firstTest` and `eachTest`. The elitist rule forbids discarding elite configurations, even if the show poor performance, until the last of the previous instances is seen in the race.

The **non-elitist** version of **irace** can discard elite configurations at any point of the race, instances are not re-used from one race to the next, and new instances are sampled for each race.

`elitistNewInstances` *flag:* `--elitist-new-instances` *default:* `1`
Number of new instances added to each race before evaluating instances from previous races (only for elitist **irace**).



If `deterministic` is `TRUE` then the number of `elitistNewInstances` will be reduced or set to `0` once all instances have been evaluated.

`elitistLimit` *flag:* `--elitist-limit` *default:* `2`
Maximum number of statistical tests performed without successful elimination after all instances from the previous race have been evaluated. If the limit is reached, the current race is stopped. Only valid for elitist **irace**. Use `0` to disable the limit.

11.3 Internal irace options

`sampleInstances` *flag:* `--sample-instances` *default:* `1`
Enable/disable the sampling of the training instances. If the option `sampleInstances` is disabled, the instances are used in the order provided in the `trainInstancesFile` or in the order they are read from the `trainInstancesDir` when`trainInstancesFile` is not provided. For more information about training instances see Section 5.4.

`softRestart` *flag:* `--soft-restart` *default:* `1`
Enable/disable the soft-restart strategy that avoids premature convergence of the probabilistic model. When a sampled configuration is *similar* to its parent configuration, the probabilistic model of these configurations is soft restarted. The soft-restart mechanism is explained in the **irace** paper [12]. The similarity of categorical and ordinal parameters is given by the hamming distance, and the option `softRestartThreshold` defines the similarity of numerical parameters.

`softRestartThreshold` *flag:* `--soft-restart-threshold` *default:* `1e-04`
Soft restart threshold value for numerical parameters.

`nbIterations` *flag:* `--iterations` *default:* `0`
Maximum number of iterations to be executed. Each iteration involves the generation of new configurations and the use of racing to select the best configurations. By default (with 0), **irace** calculates a *minimum* number of iterations as $N^{\text{iter}} = \lfloor 2 + \log_2 N^{\text{param}} \rfloor$, where N^{param} is the number of non-fixed parameters to be tuned. Setting this parameter may make **irace** stop sooner than it should without using all the available budget. We recommend to use the default value.

`nbExperimentsPerIteration` *flag:* `--experiments-per-iteration` *default:* `0`
Number of runs of the target algorithm per iteration. By default (when equal to 0), this value changes for each iteration and depends on the iteration index and the remaining budget. Further details are provided in the **irace** paper [12]. We recommend to use the default value.

`minNbSurvival` *flag:* `--min-survival` *default:* `0`
Minimum number of configurations needed to continue the execution of each race (iteration).

If the number of configurations alive in the race is not larger than this value, the current iteration will stop and a new iteration will start, even if there is budget left to continue the current race. By default (when equal to 0), the value is calculated automatically as $\lfloor 2 + \log_2 N^{\text{param}} \rfloor$, where N^{param} is the number of non-fixed parameters to be tuned.

`nbConfigurations` *flag:* `--num-configurations` *default:* `0`
The number of configurations that will be raced at each iteration. By default (when equal to 0), this value changes for each iteration and depends on `nbExperimentsPerIteration`, the iteration index and `mu`. The precise details are given in the **irace** paper [12]. We recommend to use the default value.

`mu` *flag:* `--mu` *default:* `5`
Parameter used to define the number of configurations to be sampled and evaluated at each iteration. The number of configurations will be calculated such that there is enough budget in each race to evaluate all configurations on at least $\mu + \min(5, j)$ training instances, where j is the index of the current iteration. The value of μ will be adjusted to never be lower than the value of `firstTest`. We recommend to use the default value and, if needed, adjust `firstTest`and `eachTest`, instead.

11.4 Target algorithm parameters

`parameterFile` *flag:* `-p` *or* `--parameter-file` *default:* `./parameters.txt`
File that contains the description of the parameters of the target algorithm. See Section 5.1.

11.5 Target algorithm execution

`targetRunner` *flag:* `--target-runner` *default:* `./target-runner`
Executable or R function that evaluates a configuration of the target algorithm on a particular instance. See Section 5.2 for details.

`targetRunnerLauncher` *flag:* `--target-runner-launcher` *default:*
Executable that will be used to launch the target runner, when `targetRunner` cannot be executed directly (e.g., a Python script in Windows).

`targetCmdline` *flag:* `--target-cmdline` *default:* `{configurationID} {instanceID} {seed} {instance} {bound} {targetRunnerArgs}`
Command-line arguments provided to `targetRunner` (or `targetRunnerLauncher` if defined). The substrings `{configurationID}`, `{instanceID}`, `{seed}`, `{instance}`, and `{bound}` will be replaced by their corresponding values. The substring `{targetRunnerArgs}` will be replaced by the concatenation of the switch and value of all active parameters of the particular configuration being evaluated. The substring `{targetRunner}`, if present, will be replaced by the value of `targetRunner` (useful when using `targetRunnerLauncher`). Example:

```
targetRunner="./real_target_runner.py"
targetRunnerLauncher="python"
targetCmdLine="-m {targetRunner} {configurationID} {instanceID}\
 --seed {seed} -i {instance} --cutoff {bound} {targetRunnerArgs}"
```

`targetRunnerRetries` *flag:* `--target-runner-retries` *default:* `0`
Number of times to retry a call to `targetRunner` if the call failed.

`targetRunnerTimeout` *flag:* `--target-runner-timeout` *default:* `0`
Timeout in seconds of any `targetRunner` call (only applies to `target-runner` executables not to R functions), ignored if 0.

`targetRunnerData` *default:*
Optional data passed to `targetRunner`. This is ignored by the default `targetRunner` function, but it may be used by custom `targetRunner` functions to pass persistent data around.

`targetRunnerParallel` *default:*
Optional R function to provide custom parallelization of `targetRunner`. See Section 6 for more information.

`targetEvaluator` *flag:* `--target-evaluator` *default:*
Optional script or R function that returns a numerical value for an experiment after all configurations have been executed on a given instance using `targetRunner`. See Section 5.3 for details.

`deterministic` *flag:* `--deterministic` *default:* `0`
Enable/disable deterministic target algorithm mode. If the target algorithm is deterministic, configurations will be evaluated only once per instance. See Section 5.4 for more information.


If the number of instances provided is less than the value specified for the option `firstTest`, no statistical test will be performed.

`parallel` *flag:* `--parallel` *default:* `0`
Number of calls of the `targetRunner` to execute in parallel. Values 0 or 1 mean no parallelization. For more information on parallelization, see Section 6.

`loadBalancing` *flag:* `--load-balancing` *default:* `1`
Enable/disable load-balancing when executing experiments in parallel. Load-balancing makes better use of computing resources, but increases communication overhead. If this overhead is large, disabling load-balancing may be faster. See Section 6.

`mpi` *flag:* `--mpi` *default:* `0`
Enable/disable use of **Rmpi** to execute the `targetRunner` in parallel using MPI protocol. When `mpi` is enabled, the option `parallel` is the number of slave nodes. See Section 6.

`batchmode` *flag:* `--batchmode` *default:* `0`
Specify how irace waits for jobs to finish when `targetRunner` submits jobs to a batch cluster: `sge`, `pbs`, `torque`, `slurm` or `htcondor` (`targetRunner` must submit jobs to the cluster using. for example, `qsub`). See Section 6.

11.6 Initial configurations

`configurationsFile` *flag:* `--configurations-file` *default:*
File containing a table of initial configurations. If empty or `NULL`, **irace** will not use initial configurations. See Section 5.5.


The provided configurations must not violate the constraints described in `parameterFile` and `forbiddenFile`.

11.7 Training instances

`trainInstancesDir` *flag:* `--train-instances-dir` *default:*
Directory where training instances are located; either absolute path or relative to current directory. See Section 5.4.

`trainInstancesFile` *flag:* `--train-instances-file` *default:*
File that contains a list of instances and optionally additional parameters for them. See Section 5.4.



The list of instances in `trainInstancesFile` is interpreted as file-system paths relative to `trainInstancesDir`. When using an absolute path or instances that are not files, set `trainInstancesDir=""`.

`blockSize` *flag:* `--block-size` *default:* `1`
Number of training instances, that make up a 'block' in `trainInstancesFile`. Elimination of configurations will only be performed after evaluating a complete block and never in the middle of a block. Each block typically contains one instance from each instance class (type or family) and the block size is the number of classes. The value of `blockSize` will multiply `firstTest`, `eachTest` and `elitistNewInstances`.

11.8 Tuning budget

`maxExperiments` *flag:* `--max-experiments` *default:* `0`
The maximum number of runs (invocations of `targetRunner`) that will be performed. It determines the maximum budget of experiments for the tuning. See Section 10.1.

`minExperiments` *flag:* `--min-experiments` *default:*
The minimum number of runs (invocations of `targetRunner`) that will be performed. If this option is set, then `maxExperiments` is ignored and the actual budget will depend on the number of parameters and `minNbSurvival`, but it will not be smaller than this value. See Section 10.1.

`maxTime` *flag:* `--max-time` *default:* `0`
The maximum total time for the runs of `targetRunner` that will be performed. The mean execution time of each run is estimated in order to calculate the maximum number of experiments (see option `budgetEstimation`). When `maxTime` is positive, then `targetRunner` **must** return the execution time as its second output. This value and the one returned by `targetRunner` must use the same units (seconds, minutes, iterations, evaluations, ...). See Section 10.1.

`budgetEstimation` *flag:* `--budget-estimation` *default:* `0.05`
Fraction (smaller than 1) of the budget used to estimate the mean execution time of a configuration. Only used when `maxTime` > 0. See Section 10.1.

`minMeasurableTime` *flag:* `--min-measurable-time` *default:* `0.01`
Minimum time unit that is still (significantly) measureable.

11.9 Statistical test

`testType` *flag:* `--test-type` *default:*
Specifies the statistical test used for elimination:

`F-test` (Friedman test)

`t-test` (pairwise t-tests with no correction)

`t-test-bonferroni` (t-test with Bonferroni's correction for multiple comparisons)

`t-test-holm` (t-test with Holm's correction for multiple comparisons).

We recommend to not use corrections for multiple comparisons because the test typically becomes too strict and the search stagnates. See Section 10.7 for details about choosing the statistical test most appropriate for your scenario.



The default setting of `testType` is `F-test` unless the `capping` option is enabled in which case, the default setting is `t-test`.

`firstTest` *flag:* `--first-test` *default:* `5`
Specifies how many instances are evaluated before the first elimination test.



The value of `firstTest` must be a multiple of `eachTest`.

`eachTest` *flag:* `--each-test` *default:* `1`
Specifies how many instances are evaluated between elimination tests.

`confidence` *flag:* `--confidence` *default:* `0.95`
Confidence level for the elimination test.

11.10 Adaptive capping

`capping` *flag:* `--capping` *default:*
Enable the use of adaptive capping. Capping is enabled by default if `elitist` is active, `maxTime` > 0 and `boundMax` > 0. When using this option, **irace** provides an execution bound to each target algorithm execution (See Section 5.2). For more details about this option See Section 10.4.

`cappingAfterFirstTest` *flag:* `--capping-after-first-test` *default:* `0`
If set to 1, elimination due to capping only happens after `firstTest` instances are seen.

`cappingType` *flag:* `--capping-type` *default:* `median`
Specifies the measure used to define the execution bound:

`median` (the median of the performance of the elite configurations)

`mean` (the mean of the performance of the elite configurations)

`best` (the best performance of the elite configurations)

`worst` (the worst performance of the elite configurations).

`boundType` *flag:* `--bound-type` *default:* `candidate`
Specifies how to calculate the performance of elite configurations for the execution bound:

`candidate` (performance of candidates is aggregated across the instances already executed)

`instance` (performance of candidates on each instance).

`boundMax` *flag:* `--bound-max` *default:* `0`
Maximum execution bound for `targetRunner`. It must be specified when capping is enabled.

`boundDigits` *flag:* `--bound-digits` *default:* `0`
Precision used for calculating the execution time. It must be specified when capping is enabled.

`boundPar` *flag:* `--bound-par` *default:* `1`
Penalty used for PARX. This value is used to penalize timed out executions by multiplying `boundMax`, see Section 10.4.

`boundAsTimeout` *flag:* `--bound-as-timeout` *default:* `1`
Replace the configuration cost of bounded executions with `boundMax`. See Section 10.4.

11.11 Recovery

`recoveryFile` *flag:* `--recovery-file` *default:*
Previously saved **irace** log file that should be used to recover the execution of **irace**; either absolute path or relative to the current directory. If empty or `NULL`, recovery is not performed. For more details about recovery, see Section 8.

11.12 Testing

`--only-test` *flag:* `--only-test` *default:*
Run the configurations contained in the file provided as argument on the test instances. See Section 7.

`testInstancesDir` *flag:* `--test-instances-dir` *default:*
Directory where testing instances are located, either absolute or relative to current directory.

`testInstancesFile` *flag:* `--test-instances-file` *default:*
File containing a list of test instances and optionally additional parameters for them.

`testNbElites` *flag:* `--test-num-elites` *default:* `1`
Number of elite configurations returned by irace that will be tested if test instances are provided. For more information about the testing, see Section 7.

`testIterationElites` *flag:* `--test-iteration-elites` *default:* `0`
Enable/disable testing the elite configurations found at each iteration.

12 FAQ (Frequently Asked Questions)

12.1 Is irace minimizing or maximizing the output of my algorithm?

By default, **irace** considers that the value returned by `targetRunner` (or by `targetEvaluator`, if used) should be **minimized**. In case of a maximization problem, one can simply multiply the value by -1 before returning it to irace. This is done, for example, when maximizing the hypervolume (see the last lines in `$IRACE_HOME/examples/hypervolume/target-evaluator`).

12.2 Are experiments with irace reproducible?

Short answer: Yes, under some conditions.
Long answer: According to the terminology described by López-Ibáñez et al. [13], we define *repeatability* as "*exactly repeating the original experiment, generating precisely the same results*". Following this definition, a run of **irace** is repeatable under the following conditions:

- Same version of **irace**.
- Same version of R (different versions of R may change the behavior of functions used by **irace**).
- The behavior of `targetRunner` is deterministic or exactly reproducible for the same instance, parameter configuration and random seed. Make sure that `targetRunner` uses the seed provided by **irace** to initialize all random number generators used. If the result of `targetRunner` depends on CPU-time, wall-clock time or system load in any way, then `targetRunner` is not reproducible and neither will be **irace**.
- Same random seed (`seed`) given to **irace**.
- Same scenario options (Section 11). Although some options should not affect reproducibility (e.g., `debugLevel`), maintaining a list of such options will be a huge effort, thus the safest assumption is that any change may break reproducibility.
- Same parameter space (Section 5.1), including types, domains, conditions and forbidden configurations. The order of the parameters may also affect reproducibility (the name of the parameters should not) because it affects the order in which random numbers are used.
- Same training instances provided and in the same order (Section 5.4). Even if the instances are sampled randomly (`sampleInstances`), a different initial order will produce a different sample even with the same random seed.
- Same initial configurations (Section 5.5), if any.

12.3 Is it possible to configure a MATLAB algorithm with irace?

Definitely. There are three main ways to achieve this:

1. Edit the `targetRunner` script to call MATLAB in a non-interactive way. See the MATLAB documentation, or the following links.[8] You would need to pass the parameter received by `targetRunner` to your MATLAB script.[9] There is a minimal example in `$IRACE_HOME/examples/matlab/`, where you can find an example `target-runner` script calling the `matlab` executable.
2. Call MATLAB code directly from R using the **matlabr** package (`https://cran.r-project.org/package=matlabr`). This is a better option if you are experienced in R. Define `targetRunner` as an R function instead of a path to a script. The function should call your MATLAB code with appropriate parameters. There is a minimal example in `$IRACE_HOME/examples/matlabr/`, where you can find the `targetRunner` function in `scenario.txt` and the MATLAB code in `Main.m`.

[8] See suppress-start-message-of-matlab (stackoverflow.com) and how-to-call-matlab-from-command-line-and-print-to-stdout-before-exiting (stackoverflow.com)
[9] See how-can-i-pass-input-parameters-when-running-matlab-in-batch-mode-in-windows (mathworks.com) and how-can-i-pass-command-line-arguments-to-a-standalone-matlab-executable-running (mathworks.com)

3. Another possibility is calling MATLAB directly from a different programming language and write `targetRunner` in that programming language, for example, in Python (see examples in `$IRACE_HOME/examples/target-runner-python/`).[10]

12.4 My program works perfectly on its own, but not when running under irace. Is irace broken?

Every time this was reported, it was a difficult-to-reproduce bug, i.e., a Heisenbug, in the program (target algorithm), not in **irace**. To detect such bugs, we recommend that you use, within `targetRunner`, a memory debugger (e.g., `valgrind`) to run your program. For example, if your program is executed by `targetRunner` as:

```
${EXE} ${FIXED_PARAMS} -i ${INSTANCE} ${CONFIG_PARAMS} 1> ${STDOUT} 2> ${STDERR}
```

then replace that line with:

```
valgrind --error-exitcode=1 ${EXE} ${FIXED_PARAMS} -i ${INSTANCE} \
  ${CONFIG_PARAMS} 1> ${STDOUT} 2> ${STDERR}
```

If there are bugs in your program, they will appear in `$STDERR`, thus do not delete those files. Memory debuggers will significantly slowdown your code, so use them only as a means to find what is wrong with your target algorithm. Once you have fixed the bugs, you should remove the use of `valgrind`.

12.5 irace seems to run forever without any progress, is this a bug?

Every time this problem was reported, the issue was in the target algorithm and not in **irace**. Some ideas for debugging this problem:

- Check that the target algorithm is really not running nor paused nor sleeping nor waiting for input-output.
- Use `debugLevel=3` to see how **irace** calls `target-runner`, run the same command outside **irace** and verify that it terminates.
- Add some output to your algorithm that reports at the very end the runtime and exit code. Verify that this output is printed when **irace** calls your algorithm.
- In `target-runner`, print something to a log file *after* calling your target algorithm. Verify that this output appears in the log file when **irace** is running.
- Set a maximum timeout when calling your target algorithm from `target-runner` (see FAQ 12.6).

12.6 My program may be buggy and run into an infinite loop. Is it possible to set a maximum timeout?

We are not aware of any way to achieve this using R. However, in GNU/Linux, it is easy to implement by using the `timeout` command[11] in `targetRunner` when invoking your program.

[10] See call-matlab-functions-from-python (mathworks.com) and call-user-script-and-function-from-python (mathworks.com)

[11] `https://man7.org/linux/man-pages/man1/timeout.1.html`

12.7 When using the `mpi` option, irace stops with an error message indicating that a function is not defined. How to fix this?

Rmpi does not work the same way when called from within a package and when called from a script or interactively. When **irace** creates the slave nodes, the slaves will load a copy of **irace** automatically. If the slave nodes are on different machines, they must have **irace** installed. If **irace** is not installed system-wide, R needs to be able to find **irace** on the slave nodes. This is usually done by setting `R_LIBS`, `.libPaths()` or by loading **irace** using `library()` or `require()` with the argument "`lib.loc`". The settings on the master are not applied to the slave nodes automatically, thus the slave nodes may need their own settings. After spawning the slaves, it is too late to modify those settings, thus modifying the shell variable `R_LIBS` seems the only valid way to tell the slaves where to find **irace**.

If the path is set correctly and the problem persists, please check these instructions:

1. Test that **irace** and **Rmpi** work. Run **irace** on a single machine (submit node), without calling `qsub`, `mpirun` or a similar wrapper around **irace** or R.

2. Test loading **irace** on the slave nodes. However, jobs submitted by `qsub`/`mpirun` may load R packages using a different mechanism from the way it happens if you log directly into the node (e.g., with `ssh`). Thus, you need to write a little R program such as:

   ```
   library(Rmpi)
   mpi.spawn.Rslaves(nslaves = 10)
   paths <- mpi.applyLB(1:10, function(x) {
     library(irace); return(path.package("irace")) })
   print(paths)
   ```

   Submit this program to the cluster like you would submit **irace** (using `qsub`, `mpirun` or whatever program is used to submit jobs to the cluster).

3. In the script `bin/parallel-irace-mpi`, the function `irace_main()` creates an MPI job for our cluster. You may need to speak with the admin of your cluster and ask them how to best submit a job for MPI. There may be some particular settings that you need. **Rmpi** normally creates log files; but **irace** suppresses those files unless `debugLevel` > 0.

Please contact us (Section 13) if you have further problems.

12.8 Error: 4 arguments passed to `.Internal(nchar)` which requires 3

This is a bug in R 3.2.0 on Windows. The solution is to update your version of R.

12.9 Warning: `In read.table(filename, header = TRUE, colClasses = "character", : incomplete final line found by ...`

This is a warning given by R when the last line of an input file does not finish with the newline character. The warning is harmless and can be ignored. If you want to suppress it, just open the file and press the `ENTER` key at the end of the last line of the file to end the final line with a newline.

12.10 How are relative filesystem paths interpreted by irace?

The answer depends on where the path appears. Relative paths may appear as the argument of command-line options, as the value of options given in the scenario file, or within various scripts, functions or instance files. Table 1 summarizes how paths are translated from relative to absolute.

Table 1: Translation of relative to absolute filesystem paths.

Relative path appears as ...	**... is relative to ...**
a string within `trainInstancesFile`	`trainInstancesDir`
a string within `testInstancesFile`	`testInstancesDir`
code within `targetRunner` or `targetEvaluator`	`execDir`
the value of `logFile` or `--log-file`	`execDir`
the value of other options in the scenario file	the directory containing the scenario file
the value of other command-line options	invocation (working) directory of **irace**

12.11 My parameter space is small enough that irace could generate all possible configurations; however, irace generates repeated configurations and/or does not generate some of them. Is this a bug?

Typically, **irace** is applied to parameter spaces that are much larger than what can be explored within the budget given. Thus, **irace** does not try to detect whether all possible configurations can be evaluated for the given budget and it does not waste computation time to check for repeated configurations. Thus, if the parameter space is actually very small, the initial random sampling performed by **irace** may generate repeated configurations and/or never generate some configurations, which is not ideal. If you still want to use (non-iterated) racing, the recommended approach is to provide all configurations explicitly to **irace** (Section 5.5) and execute a single race (`nbIterations=1`) with exactly the number of configurations provided (e.g., `nbConfigurations=240`). A future version of **irace** may automatically detect this case and switch to non-iterated racing without having to set additional options. Future versions may also implement computationally cheap checks for repeated configurations.[12]

12.12 On Windows and using `target-runner.py` (a **Python** file), I get the error "`target-runner.py is not executable`"

The issue is that `.py` files are not executable on their own and you need `python.exe` to read the `.py` file and execute it. Linux knows how to do this if the first line of the file is "`#!/usr/bin/python`", however, Windows doesn't know how to do it. In Windows you have 2 options:

- Create a `target-runner.bat` file that contains a line similar to (see `templates/windows/target-runner.bat`):

```
C:\path\to\python.exe C:\path\to\target-runner.py %instance% %seed% \
  %candidate_parameters% 1>%stdout% 2>%stderr%
```

[12] If you are interested in implementing this, please contact us!

- Or convert `target-runner.py` into an `.exe` file, for example, using `auto-py-to-exe`[13], so that you do not need a `.bat` file.

12.13 Error in `socketConnection("localhost", port = port, server = TRUE, lock = TRUE, : can not open the connection`

This error may arise if you activate the `parallel` option of **irace** and your `targetRunner` or `targetEvaluator` tries to setup a parallel cluster or execute code in parallel in a way that interacts badly with the parallel mechanism in R. In this case, you need to either investigate yourself if there is a way for the two parallel mechanisms to co-exist or, if that is not possible, disable parallelism in **irace** or in your code. Note that packages or software used by your `targetRunner` may have a parallel mechanism enabled by default and unknown to you. This is definitely NOT a bug in **irace**.

12.14 irace does not print the call to the `targetRunner` with `debugLevel`=2 when using the `parallel`

This is a limitation of Windows or Rstudio. Running without `parallel` should work. Unfortunately, we cannot fix this limitation in **irace**. If you need to understand how **irace** calls `targetRunner` when running in parallel, you can implement a logging mechanism able to handle parallelism directly inside the `targetRunner`.

12.15 Why does the column `Bound` in the output sometimes shows `NA`?

This should only happen when the symbol in the first column is '`.`', which means that all alive configurations are elite and have already been evaluated on this instance, so nothing is evaluated nor discarded.

13 Resources and contact information

More information about the package can be found on the **irace** webpage:

https://mlopez-ibanez.github.io/irace/

For questions and suggestions please contact the development team through the **irace** package Google group:

https://groups.google.com/d/forum/irace-package

or by sending an email to:

irace-package@googlegroups.com

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[13] https://pypi.org/project/auto-py-to-exe/

Bibliography

[1] A. Biedenkapp, M. T. Lindauer, K. Eggensperger, F. Hutter, C. Fawcett, and H. H. Hoos. Efficient parameter importance analysis via ablation with surrogates. In S. P. Singh and S. Markovitch, editors, *Proceedings of the AAAI Conference on Artificial Intelligence*. AAAI Press, Feb. 2017. doi: 10.1609/aaai.v31i1.10657.

[2] M. Birattari. On the estimation of the expected performance of a metaheuristic on a class of instances. how many instances, how many runs? Technical Report TR/IRIDIA/2004-001, IRIDIA, Université Libre de Bruxelles, Belgium, 2004.

[3] M. Birattari, Z. Yuan, P. Balaprakash, and T. Stützle. F-race and iterated F-race: An overview. In T. Bartz-Beielstein, M. Chiarandini, L. Paquete, and M. Preuss, editors, *Experimental Methods for the Analysis of Optimization Algorithms*, pages 311–336. Springer, Berlin/Heidelberg, 2010. doi: 10.1007/978-3-642-02538-9_13.

[4] M. De Souza, M. Ritt, M. López-Ibáñez, and L. Pérez Cáceres. ACVIZ: A tool for the visual analysis of the configuration of algorithms with irace. *Operations Research Perspectives*, 8: 100186, 2021. doi: 10.1016/j.orp.2021.100186.

[5] M. De Souza, M. Ritt, and M. López-Ibáñez. Capping methods for the automatic configuration of optimization algorithms. *Computers & Operations Research*, 139:105615, 2022. doi: 10.1016/j.cor.2021.105615.

[6] C. Fawcett and H. H. Hoos. Analysing differences between algorithm configurations through ablation. *Journal of Heuristics*, 22(4):431–458, 2016. doi: 10.1007/s10732-014-9275-9.

[7] F. Hutter, H. H. Hoos, K. Leyton-Brown, and T. Stützle. ParamILS: an automatic algorithm configuration framework. *Journal of Artificial Intelligence Research*, 36:267–306, Oct. 2009. doi: 10.1613/jair.2861.

[8] F. Hutter, H. H. Hoos, and K. Leyton-Brown. Identifying key algorithm parameters and instance features using forward selection. In P. M. Pardalos and G. Nicosia, editors, *Learning and Intelligent Optimization, 7th International Conference, LION 7*, volume 7997 of *Lecture Notes in Computer Science*, pages 364–381. Springer, Heidelberg, Germany, 2013. doi: 10.1007/978-3-642-44973-4_40.

[9] F. Hutter, H. H. Hoos, and K. Leyton-Brown. An efficient approach for assessing hyperparameter importance. In E. P. Xing and T. Jebara, editors, *Proceedings of the 31st International Conference on Machine Learning, ICML 2014*, volume 32, pages 754–762. PMLR, 2014. URL https://proceedings.mlr.press/v32/hutter14.html.

[10] M. López-Ibáñez and T. Stützle. Automatically improving the anytime behaviour of optimisation algorithms. *European Journal of Operational Research*, 235(3):569–582, 2014. doi: 10.1016/j.ejor.2013.10.043.

[11] M. López-Ibáñez, J. Dubois-Lacoste, T. Stützle, and M. Birattari. The irace package, iterated race for automatic algorithm configuration. Technical Report TR/IRIDIA/2011-004, IRIDIA, Université Libre de Bruxelles, Belgium, 2011. URL https://permalink.ulb.be/t/JFgRQ92OZl/download. Published in Operations Research Perspectives [12].

[12] M. López-Ibáñez, J. Dubois-Lacoste, L. Pérez Cáceres, T. Stützle, and M. Birattari. The irace package: Iterated racing for automatic algorithm configuration. *Operations Research Perspectives*, 3:43–58, 2016. doi: 10.1016/j.orp.2016.09.002.

[13] M. López-Ibáñez, J. Branke, and L. Paquete. Reproducibility in evolutionary computation. *ACM Transactions on Evolutionary Learning and Optimization*, 1(4):1–21, 2021. doi: 10.1145/3466624.

[14] M. López-Ibáñez, D. Vermetten, J. Dreo, and C. Doerr. Using the empirical attainment function for analyzing single-objective black-box optimization algorithms. *IEEE Transactions on Evolutionary Computation*, 29(5):1774–1782, 2025. doi: 10.1109/TEVC.2024.3462758.

[15] C. C. McGeoch. Analyzing algorithms by simulation: Variance reduction techniques and simulation speedups. *ACM Computing Surveys*, 24(2):195–212, 1992. doi: 10.1145/130844.130853.

[16] L. Pérez Cáceres, M. López-Ibáñez, H. H. Hoos, and T. Stützle. An experimental study of adaptive capping in irace. In R. Battiti, D. E. Kvasov, and Y. D. Sergeyev, editors, *Learning and Intelligent Optimization, 11th International Conference, LION 11*, volume 10556 of *Lecture Notes in Computer Science*, pages 235–250. Springer, Cham, Switzerland, 2017. doi: 10.1007/978-3-319-69404-7_17.

[17] M. Schneider and H. H. Hoos. Quantifying homogeneity of instance sets for algorithm configuration. In Y. Hamadi and M. Schoenauer, editors, *Learning and Intelligent Optimization, 6th International Conference, LION 6*, volume 7219 of *Lecture Notes in Computer Science*, pages 190–204. Springer, Heidelberg, Germany, 2012. doi: 10.1007/978-3-642-34413-8_14.

[18] Z. Yuan, M. A. Montes de Oca, T. Stützle, H. C. Lau, and M. Birattari. An analysis of post-selection in automatic configuration. In C. Blum and E. Alba, editors, *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2013*, pages 1557–1564. ACM Press, New York, NY, 2013. ISBN 978-1-4503-1963-8.

Appendix A Installing R

This section gives a quick R installation guide that will work in most cases. The official instructions are available at https://cran.r-project.org/doc/manuals/r-release/R-admin.html

A.1 GNU/Linux

You should install R from your package manager. On a Debian/Ubuntu system it will be something like:

```
sudo apt-get install r-base
```

Once R is installed, you can launch R from the Terminal and from the R prompt install the **irace** package (see Section 3.2).

A.2 OS X

You can install R directly from a CRAN mirror.[14] Alternatively, if you use homebrew, you can just brew the R formula from the science tap (unfortunately it does not come already bottled so you need to have Xcode[15] installed to compile it):

```
brew tap homebrew/science
brew install r
```

Once R is installed, you can launch R from the Terminal (or from your Applications), and from the R prompt install the **irace** package (see Section 3.2).

A.3 Windows

You can install R from a CRAN mirror.[16] We recommend that you install R on a filesystem path without spaces, special characters or long names, such as `C:\R`. Once R is installed, you can launch the R console and install the **irace** package from it (see Section 3.2).

Appendix B targetRunner troubleshooting checklist

If the `targetRunner` script fails to return the output expected by **irace**, it can be sometimes difficult to diagnose where the problem lies. The more descriptive errors provided by your script, the easier it will be to debug it. If `targetRunner` enters an infinite loop, irace will wait indefinitely (see FAQ in Section 12.6). If you are using temporary files to redirect the output of your algorithm, check that these files are properly created. We recommend to follow the structure of the example file (`target-runner`) provided in `$IRACE_HOME/templates`. The following error examples are based on that example file.

In case of failure of `targetRunner`, **irace** will print an error on its output describing which execution of `targetRunner` failed. Follow this checklist to detect where the problem is:

1. Make sure that your `targetRunner` script or program is at the specified location. If you see this error:

[14] https://cran.r-project.org/bin/macosx/
[15] Xcode download webpage: https://developer.apple.com/xcode/download/
[16] https://cran.r-project.org/bin/windows/

```
Error: == irace == target runner './tuning/target-runner' does not exist
```

it means that **irace** cannot find the `target-runner` file. Check that the file is at the path specified by the error.

2. Make sure that your `targetRunner` script is an executable file and the user running **irace** has permission to execute it. The following errors:

```
Error: == irace == target runner './tuning/target-runner' is a directory,
not a file
```

or

```
Error: == irace == target runner './tuning/target-runner' is not executable
```

mean that your `targetRunner` is not an executable file. In the first case, the script is a folder and therefore there must be a problem with the name of the script. In the second case, you must make the file executable, which in GNU/Linux can be done by:

```
chmod +x ./tuning/target-runner
```

3. If your `targetRunner` script calls another program, make sure it is at the location described in the script (variable `EXE` in the examples and templates). A typical output for such an error is:

```
Error: == irace == running command ''./tuning/target-runner' 1 8 676651103
./tuning/Instances/1000-16.tsp --ras --localsearch 2 --alpha 4.03 --beta 1.89
--rho  0.02 --ants 37 --nnls 48 --dlb 0 --rasranks 15 2>\&1' had status 1
== irace == The call to target_runner_default was:
./tuning/target-runner 1 8 676651103 ./tuning/Instances/1000-16.tsp --ras
--localsearch 2 --alpha 4.03 --beta 1.89 --rho  0.02 --ants 37 --nnls 48
--dlb 0 --rasranks 15
== irace == The output was:
Tue May  3 19:00:37 UTC 2016: error: ./bin/acotsp: not found or not executable
(pwd: ./tuning/acotsp-arena)
```

You may test your script by copying the command line shown in the error and executing `target-runner` directly on the execution directory (`execDir`). In this case, the command line is:

```
./tuning/target-runner 1 8 676651103 ./tuning/Instances/1000-16.tsp --ras \
 --localsearch 2 --alpha 4.03 --beta 1.89 --rho 0.02 --ants 37 --nnls 48 \
 --dlb 0 --rasranks 15
```

This executes the `targetRunner` script as **irace** does. The output of this script must be only one number.

4. If your `targetRunner` file is an executable script in Python, R, Perl, Bash or some other interpreted programming language, make sure that the interpreter specified in the first line of the file exists at the correct location. For example, if the first line of `target-runner.py` is:

```
#|/usr/bin/python
```

Then make sure that `/usr/bin/python` exists and it is executable. Otherwise, you will get an error such as:

```
Error: == irace == error in running command
```

5. Check that your `targetRunner` script is actually returning one number as output. For example:

```
Error: == irace == The output of './tuning/target-runner 1 25 365157769
 ./tuning/Instances/1000-31.tsp --ras --localsearch 1 --alpha 0.26 --beta
 6.95 --rho 0.69 --ants 56 --nnls 10 --dlb 0 --rasranks 7' is not numeric!
== irace == The output was:
Solution: 24479793
```

   In the example above, the output of `target-runner` is "`Solution:  24479793`", which is not a number. If `target-runner` is parsing the output of the target algorithm, you need to verify that the code only parses the solution cost value.

6. Check that your `targetRunner` script is creating the output files for your algorithm. If you see an error as:

```
== irace == The output was: Tue May  3 19:41:40 UTC 2016:
error: c1-9.stdout: No such file or directory
```

   The output file of the execution of your algorithm has not been created (check permissions) or has been deleted before the result can be read.

7. Other errors can produce the following output:

```
== irace == The output was: Tue May  3 19:49:06 UTC 2016:
error: c1-23.stdout: Output is not a number
```

   This might be because your `targetRunner` script is not executing your algorithm correctly. To further investigate this issue, comment out the line that eliminates the temporary files that saves the output of your algorithm. Similar to this one

```
rm -f "${STDOUT}" "${STDERR}"
```

   Execute directly the `targetRunner` command-line that is provided in the error message, look in your execution directory for the files that are created. Check the `.stderr` file for errors and the `.stdout` file to see the output that your algorithm produces.

8. Some command within `targetRunner` may not be working correctly. In that case, you must debug the commands individually exactly as **irace** executes them. In order to find where the problem is, print the commands to a log file before executing them. For example:

```
echo "$EXE ${FIXED_PARAMS} -i $INSTANCE ${CONFIG_PARAMS}" >> ${STDERR}.log
$EXE ${FIXED_PARAMS} -i $INSTANCE ${CONFIG_PARAMS} 1> ${STDOUT} 2> ${STDERR}
```

   then look at the `$STDERR.log` file corresponding to the `targetRunner` call that failed and execute/debug the last command there.

9. If the language of your operating system, the `target-runner` or the target algorithm is not English, **irace** may not be able to recognize the numbers generated by `target-runner`. We recommend that you run **irace**, the `target-runner` and the target algorithm under an English locale (or make sure that their languages and number format are compatible).

10. It is possible that transient bugs in the target algorithm are only visible when running within **irace**, and all commands within `targetRunner` appear to work fine when executed directly in the command-line outside **irace**. See FAQ in Section 12.4) for suggestions on how to detect such bugs.

11. If your `targetRunner` script works when running irace with `parallel=0` but it fails when using higher number of cores, this may be due to any number of reasons:

    - If you submit jobs through a queuing system, the running environment when using the queuing system may not be the same as when you launch **irace** yourself. The queuing system may also send the job to different machines depending on the number of CPUs requested. One way to test this is to submit the failing execution of `targetRunner` to the queuing system, and specifically to any problematic machine.
    - When using MPI, some calls to `targetRunner` may run on different computers than the one running the master **irace** process. See FAQ in Section 12.7.
    - Does `targetRunner` read or create intermediate files? These files may cause a race condition when two calls to `targetRunner` happen at the same time. You have to make sure that parallel runs of `targetRunner` do not interfere with each other's files.
    - Maybe these files consume too much memory or fill the filesystem when there are simultaneous `targetRunner` calls? Moreover, queuing systems have stricter limits for computing nodes than for the submit/host node.
    - Does the machine or the queuing system impose any limits on number of processes or CPU/memory/filesystem usage per job? Such limits may only trigger when more than one process is executed in parallel, killing the `targetRunner` process before it has a chance to print anything useful. In that case, **irace** may not detect the the program finished unexpectedly, only that the expected output was not printed.

Appendix C targetEvaluator troubleshooting checklist

Even if `targetRunner` appears to work, the use of `targetEvaluator` may lead to other problems. The same checklist of `targetRunner` can be followed here. In addition, we list here other potential problems unique to `targetEvaluator`:

1. If `targetEvaluator` fails only in the second or later iteration, this may because output files or data generated by a previous call to `targetRunner` are missing. Elite configurations are never re-executed on the same instance and seed pair, that is, **irace** will call only once `targetRunner` for each pair of configuration ID and instance ID. However, `targetEvaluator` is always re-executed, which takes into account any updated information (normalization bounds, reference sets/points, best-known values, etc.). Thus, any files or data generated by `targetRunner` for a given configuration must remain available to `targetEvaluator` as long as that configuration is alive. The list of alive configurations is passed to `targetEvaluator`, which may decide then which data to keep or remove.

Appendix D Glossary

Parameter tuning: Process of searching good settings for the parameters of an algorithm under a particular tuning scenario (instances, execution time, etc.).

Scenario: Settings that define an instance of the tuning problem. These settings include the algorithm to be tuned (target), budget for the execution of the target algorithm (execution time, evaluations, iterations, etc.), set of problem instances and all the information that is required to perform the tuning.

Target algorithm: Algorithm whose parameters will be tuned.

Target parameter: Parameter of the target algorithm that will be tuned.

irace option: Configurable option of **irace**.

Elite configurations: Best configurations found so far by **irace**. New configurations for the next iteration of **irace** are sampled from the probabilistic models associated to the elite configurations. All elite configurations are also included in the next iteration.

$IRACE_HOME: The filesystem path where **irace** is installed. You can find this information by opening an R console and executing:

```
system.file(package = "irace")
```

Appendix E NEWS

NEWS

```
# irace (development version)

* Configuration inputs may omit variable parameters that are inactive in every configuration.


**If you are viewing this file on CRAN, please check [latest news on the irace website](https://mlopez-ibanez.github.io/

# irace 4.5.1

## Fixes

* Fix #103: The soft restart no longer fails with `subscript out of bounds` when a parameter with a dependent domain is
* Fix #102: `irace()` no longer fails when a parameter is named `n`, `nx`, `param` or `this_model`.  (Reported by @Saeth


# irace 4.5

## New features and improvements

 * Fix #93: Add an optional `iterationCallback` scenario function that is
   invoked after each completed irace iteration.  This allows irace users to
   monitor tuning progress or collect intermediate results without modifying
   irace internals. Check the [User Guide](https://mlopez-ibanez.github.io/irace/irace-package.pdf) for more details and
   implementation.

## Fixes

 * Fix bug when rejecting configurations.  The bug triggers a warning in R 4.7.
 * Fix #99: Sampling falls back to uniform sampling with a warning if there are
   more than 1024 parameters, which exceeds the dimension limit of the
   Sobol method.

# irace 4.4.4
```

```
 * The package contains an example of using `irace` to tune algorithms
   implemented in Julia language.

 * Fix #94: `sample_model.ParamOrd()` now samples uniformly when the model mean
   is `NA`, avoiding `NA` values for active ordinal parameters.  (Reported by @Saethox)

 * Fix #97: `psRace()` errors when the post-selection race has nothing left to
   run. Fixed by @ivchicano.

# irace 4.4.3

 * Do not run postselection race unless elitist racing is enabled.

# irace 4.4.2

 * Fix failing assert with `elitist=FALSE` (issue #90).

# irace 4.4.1

 * Minor release to skip failing test on CRAN.

# irace 4.4.0

## New features and improvements

 * The [User Guide](https://mlopez-ibanez.github.io/irace/irace-package.pdf) now contains a section on tuning for anytim

## Fixes

 * `irace` could error when tuning a single parameter and initial
   configurations are given for this single parameter in an input file.
   (Reported by Xiaolu Liu)

 * If `targetRunner` returns the expected output in separate lines, `irace`
   could error with: `Error in set(target_output, j = "configuration", value = unlist_element(experiments, "id_configura

 * irace will now call the RNG of R before evaluating a new instance.  This
   change should make no difference to well-behaved `targetRunners` that use
   the seed provided by irace to reseed the RNG, as recommended. However,
   `targetRunners` written in R that fail to follow this recommendation could
   end up with the same sequence of random numbers in different instances. Now,
   different instances should get different random numbers.


# irace 4.3

## New features and improvements

 * Updated MATLAB examples and instructions in the User Guide.

## Fixes

 * Add extra whitespace before the first and after the last boxplots in
   `plotAblation()` so that boxes will not overlap with plot margins.

 * Fix bug in `readConfigurationsFile()` when a fixed parameter that is both
   conditional and disabled does not appear in the input file.

 * Fix #87: Using a dependent parameter may produce an error when sampling a
   different parameter.
```

```
# irace 4.2.0

## Major breaking changes

 * Recovery (`--recover-file`) has been completely re-implemented.  This
   version of irace cannot recover files generated by previous versions and
   vice versa.

 * Now `blockSize` multiplies `elitistNewInstances`, which should ensure that
   irace evaluates instances in blocks.

 * Several improvements in `ablation()` may result in different ablation
   results using this version of irace. Also, the log generated by `ablation()`
   has some important changes to record more detailed information and it is
   more consistent with the log generated by `irace()`.

## New features and improvements

 * When `targetRunner` or `targetEvaluator` are implemented in R, using random
   numbers within those functions does not affect the sequence of random
   numbers within irace.

 * `psRace()` gains a `psrace_logFile` argument to avoid overwriting `scenario$logFile`.

 * `psRace()` makes better use of the budget remaining.

 * New scenario option `cappingAfterFirstTest`
   (`--capping-after-first-test`). If set to 1, elimination due to capping only
   happens after `firstTest` instances are seen (issue #78, suggested by Nguyen Dang).

 * `getConfigurationById()` now returns configurations in the same order
   (including repeated values) of the IDs passed as argument.

 * `ablation(type="full")` will evaluate all intermediate configurations on all
   instances at once.  This makes better use of parallel resources but will not
   report any information about progress after each instance.

 * `ablation()` now works better with dependent-domain parameters and with
   complicated conditions.  Ablation will not generate intermediate
   configurations that are forbidden.

 * The ACOTSP example in the user-guide now uses the ACOTSPQAP code and provides more guidance in case of errors.

## Fixes

 * Fix #79: irace.exe did not encode the `libPath` correctly in R >= 4.0 on Windows, producing 'unrecognized escape in c

 * Fix #76: Recovery (`--recover-file`) is working again with a completely new implementation.

 * Fixed documentation of `psRace()`.

 * Fix `psRace()` not saving `psrace_log` in `scenario$logFile`.

 * Fix bug in `psRace()` using more budget than the maximum given.

 * Handle the case when all elites are rejected before starting the race and
   `sum(alive) <= minSurvival`.

 * Fix bug in `sampleSobol()` with parameters whose domain depends on a conditional parameter.
```

```
# irace 4.1

## Major breaking changes

 * Recovery (`--recover-file`) is currently broken (issue #76).

 * The `experiment` list passed to the `targetRunner` R function (see
  `target_runner_default()`) does not contain an element `switches`.
  This element can be obtained from `scenario$parameters$switches`.


## New features and improvements

 * Conditions that are always true because they depend on fixed parameters are
   replaced by a `TRUE` value to speed up their evaluation. This speed-up is only
   measurable with very large and complicated parameter spaces.

 * The options `src` and `target` of `ablation()` (or `--src` and `--target` of
   the command-line `ablation` tool) now accept a character string that points
   to a file containing a configuration that will be read with
   `readConfigurationsFile()`.  This allows performing ablation between
   configurations not explored before.

## Fixes

 * Fixed some typos in the user guide. (@sbomsdorf)

 * Fixed several bugs in post-selection including `Error in if (any(left >= 0L)) { : missing value where TRUE/FALSE need

 * Fix bug in `ablation()` not calling `stop_parallel()`.


# irace 4.0

## Major breaking changes

 * Requires R version >= 4.0

 * Logfiles `*.Rdata` use format version 3, which can only be read by R version >= 3.5.

 * The scenario options `forbiddenFile` and `forbiddenExps` have been removed
   and will give an error if present.  Forbidden configurations are now
   specified in the parameter space description. See the example in
   `readParameters()`.

 * The scenario option `digits` has been removed and will give an error if
   present. The number of `digits` for real-valued parameters is now specified
   in the parameter space description. See the example in `readParameters()`.

 * The default value of the scenario option `trainInstancesDir` is now `""`.
   The previous default value of `"./Instances"` often caused confusion to
   users not using files as training instances.

 * The `scenario` object now includes the `parameters` object. Thus
   functions such as `irace()`, which previously took as arguments both
   `scenario` and `parameters`, now only take `scenario`. This also means that
   the log file `irace.Rdata` does not contain a separate `parameters` element
   since this element can now be found within `scenario`.

 * The default value of the scenario option `softRestartThreshold` is now
   0.0001 and does not depend on `digits`.
```

```
* The command-line executables `irace` and `ablation` (`irace.exe` and
  `ablation.exe` in Windows) will load the version of the `irace` package that
  is found in the same path where the executables are. In earlier versions,
  the executables will always load the version found via `base::.libPaths()`.
  This change allows installing multiple versions of the irace package in
  different locations and each executable will use its corresponding version.
  The correct location can be verified by looking at the line `"installed at:"`
  printed in the output.

* Adaptive capping is now enabled by default if `maxTime > 0` and `boundMax > 0`.
  It can be disabled with `--capping 0` in the command-line options or `capping=0` in the scenario options. See [Pérez-
                                    (Leslie Pérez Cáceres, Manuel López-Ibáñez)

* The scenario option `targetRunnerLauncherArgs`, introduced in version 3.5,
  was removed and replaced by `targetCmdline`, which is more flexible (fixes #38).
  Please see the user-guide for details.

* Command-line options in joined form, given as `"--log-file= --check"`, without
  any argument after the `'='` will be interpreted as an empty argument,
  equivalent to using `logFile=""` in `scenario.txt`.

* irace will now give an error if you try to recover from a file generated by
  a different version of irace, since such attempts typically end up in errors
  that are difficult to understand.

* irace warns about using `'&&'` and `'||'` instead of `'&'` and `'|'` in
  parameter conditions and forbidden expressions. A future version of irace
  will reject those uses as errors.

* The internal function `irace.reload.debug()` has been removed.
  Use [`devtools::reload()`](https://devtools.r-lib.org/reference/reload.html) instead.

* The column `"instance"` of the `instancesList` data frame stored in the
  logFile has been renamed to `"instanceID"`.  This data frame should not be
  accessed directly. Instead use the new function `get_instanceID_seed_pairs()`.

* Using `maxTime > 0` with `elitist=0` now gives a clear error rather than fail later.
                                                    (fix #65, reported by @DEOCH)

* `path_rel2abs()` will not expand symlinks to avoid problems with Python's venv.
                                                    (fix #64, reported by @DEOCH)

* Expansion of `'~'` in Windows now follows the definition of `fs::path_expand()` rather than `base::path.expand()`.

* irace is now more strict in enforcing runtime bounds given with `scenario$boundMax`
  and will stop with an error if the `target-runner` reports a runtime larger than the given bound.

* All functions that contained a period (`'.'`) in the name have been renamed to use `'_'` instead.

* The periods (`'.'`) in the arguments of `scenario$targetRunnerParallel`
  and `scenario$targetEvaluator` have also been replaced by `'_'`.

* The environment `.irace` that was available in the log file under
  `iraceResults$state$.irace` is replaced directly by `iraceResults$state`.
  It contains similar information but some entries have been renamed. For
  example, the `experimentLog` data frame is now called `experiment_log`
  and it is a [`data.table`](https://cran.r-project.org/package=data.table).

* The interface of `psRace()` has been simplified.

* `irace` will automatically execute a post-selection race (`psRace()`) using
  any remaining budget (currently only when `maxTime == 0`). To disable
```

```
   this behavior, set the scenario option `postselection` (`--postselection`)
   to `0`.


## New features and improvements

 * `sampleUniform()` and `sampleModel()` are significantly faster thanks to using [`data.table`](https://cran.r-project.

 * Initial configurations are sampled using Sobol low-discrepancy sequences using `spacefillr::generate_sobol_set()`. Th

 * Parameter spaces can be constructed programmatically using `parametersNew()`. See the documentation for details.

 * Ablation will report configurations that produced the same results, which
   indicates parameter values that have the same effect on the target algorithm,
   possibly indicating a bug in the target algorithm.

 * New option `instancesFile` of `ablation()` for using either the training
   instances, the test instances or reading instances from a given file.

 * New option `nrep` of `ablation()` specifies the number of replications per
   instance used in `"full"` ablation. It replaces the previous parameter
   `n_instances`, whose definition was more difficult to use correctly.

 * Matrix operations are faster thanks to `matrixStats`.

 * New scenario option `blockSize` for defining blocks of instances.
   Configurations are only eliminated after evaluating a complete block and
   never in the middle of a block. This is useful for scenarios when there are
   clearly defined instance classes and the best configuration should be
   balanced among them. In that case, `trainInstancesFile` should be written
   so that each block contains one instance of each class and
   `blockSize` is set to the number of classes.

 * New scenario option `targetRunnerTimeout`: Timeout in seconds of any
   `targetRunner` call (only applies to `target-runner` executables not to R
   functions).

 * `plotAblation()` has several new options:
    - `type='rank'` to plot ranks per instance instead of raw cost values.
    - `n` to limit the number of parameters shown in the plot.
    - `width` replaces `pdf.width`.
    - `height` sets the height of the plot in the PDF file.

 * The previously internal function `check.output.target.runner` is renamed to
   `check_output_target_runner` and exported to allow users who write their own
   `targetRunnerParallel` to check the output and repair it if possible.
   (Deyao Chen)

 * New functions `read_ablogfile()`, `has_testing_data()`, `irace_summarise()`.

 * New functions `get_random_seed()`, `set_random_seed()`,
   `restore_random_seed()` useful for writing `targetRunner` functions in R.

 * New function `get_instanceID_seed_pairs()` to get the pairs of instanceID
   and random seed used during the races (and optionally the actual instances).

 * The `parameters` object now stores the number of `digits` (decimal places
   after the point) for each parameter of type `r`. As a result, the
   `repairConfiguration` function (see `defaultScenario()`) only needs two
   arguments: `configuration` and `parameters`. See examples in the user-guide.

 * `readScenario()` (and command-line irace) do not require a `scenario.txt` file.
```

(Contributed by @DE0CH)

* `read_pcs_file()` now supports forbidden configurations.

* When testing, `irace` now prints the random seed used for each instance as an additional column.

* The package provides a new executable `target-runner-dummy` (or
  `target-runner-dummy.exe` in Windows) for the purposes of testing. It may
  also be useful for understanding the typical setup of `irace`.

* New scenario option `minExperiments` to set a minimum budget of runs.
(proposed by @Saethox, fixes #58)

* New function `multi_irace()` for executing multiple runs of irace with the
  same or different scenarios and parameters, possibly in parallel.
(Contributed by @Saethox)

Fixes

* `ablation_cmdline()` and `plotAblation()` no longer create an empty `Rplots.pdf` file when specifying an output PDF f

* Fix #66: when using `maxTime > 0`, irace estimates the time per run by
  executing 2 configurations on `firstTest` instances and adjusts `boundMax`
  to not go over `budgetEstimation`. This may result in a smaller `boundMax`
  than before. To reduce this impact, the default value of `budgetEstimation`
  is now `0.05` instead of `0.02`.
(Manuel López-Ibáñez, reported by @DE0CH)

* Fix #55: Configurations provided may use `<NA>` in addition to `NA` to denote
  the missing value of a disabled parameter.
(Manuel López-Ibáñez, reported by @TheIronBorn)

* Fix #44: irace now will give an error if the domain of real-valued (r)
  parameters would change depending on the value of `'digits'`. The solution
  is to increase the value of `'digits'` or adjust the domain.
(Manuel López-Ibáñez, reported by @mb706)

* If scenario option `targetRunnerParallel` is set, irace no longer tries to
  initialize a parallel environment or setup MPI. It is now the responsibility
  of the user to do that before calling irace or within the function assigned
  to `targetRunnerParallel`.

* irace no longer sets `option(error=utils::recover())` in debug mode to avoid issues
  when calling irace from Python. The user can set this if desired.

* Fix bug failing to restart with parameters that have dependent domains.

* Fix bug with `sampleInstances=FALSE` that could re-evaluate the same
  (instance, seed) pair more than once.

* Fix bug when using `targetRunnerLauncher` and `targetRunner` contain whitespace.

* Fix bug in `ablation_cmdline()` about missing `scenario` object.

* `ablation()` will now save and restore the previous random seed.

* `ablation()` will detect if the logfile (e.g., `irace.Rdata`) is incomplete.

* `readConfigurationsFile()` now handles parameters with dependent domains.

* Fix #71: Ensure `".ID."` is the first column in `checkTargetFiles()` (Manuel López-Ibáñez, reported by @ivan1arriola)

```
# irace 3.5

## New features and improvements

 * Handling of dependent parameter domains: These should be specified in the
   parameter domain definition and, for now, only numerical parameter can
   define dependent domains. A numerical domain can be dependent on one bound,
   e.g. `(1, "param1*2")`, where the dependent bound can include basic
   arithmetic operators.          (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The package now provides an `ablation` executable (`ablation.exe` in
   Windows) that makes easier to perform ablation analysis without having any R
   knowledge.

 * The interface to functions `ablation()` and `plotAblation()` has been
   simplified. The `ablation()` function now allows overriding scenario
   settings. The `plotAblation()` function will not create the plot if the
   ablation log does not contain a complete ablation.
                                                    (Manuel López-Ibáñez)

 * The argument `n.instances` of `ablation()` has been renamed to `n_instances`
   and it is now a factor that multiplies `scenario$firstTest`.
                                                         (Manuel López-Ibáñez)

 * New command-line option `--quiet` to run without producing any output
   except errors (also available as a scenario option).
                                                     (Manuel López-Ibáñez)

 * New command-line option `--init` to initialize a scenario. (Deyao Chen)

 * Added support for HTCondor cluster framework to `--batchmode`.
                                                     (Filippo Bistaffa)

 * `--check` now also check the contents of `configurationsFile` and runs
   configurations provided via `initConfigurations`.
                         (Manuel López-Ibáñez, reported by Andreea Avramescu)

 * New scenario options `targetRunnerLauncher` and `targetRunnerLauncherArgs`
   to help in cases where the target-runner must be invoked via another
   software with particular options (such as `python.exe` in Windows).
                                                           (Manuel López-Ibáñez)

 * New scenario option `minMeasurableTime`.
                                                     (Manuel López-Ibáñez)

 * An error is produced if a variable set in the scenario file is not known to
   irace.  If your scenario file contains R code, then use variable names
   beginning with a dot `'.'`, which will be ignored by irace.
                                                    (Manuel López-Ibáñez)

 * Plotting functions have been moved to the new package
   [iraceplot](https://auto-optimization.github.io/iraceplot/).  In particular,
   `configurationsBoxplot()` is replaced by `iraceplot::boxplot_training()` and
   `iraceplot::boxplot_test()`; `parallelCoordinatesPlot()` is replaced by
   `iraceplot::parallel_cat()` and `iraceplot::parallel_coord()`; and
   `parameterFrequency()` is replaced by `iraceplot::sampling_frequency()`.
                                    (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The user-guide now contains a detailed section on "Hyper-parameter
   optimization of machine learning methods".
                                                    (Manuel López-Ibáñez)
```

```
 * When `testType="F-test"` and only two configurations remain, the elimination
   test now uses the pseudo-median estimated by the Wilcoxon signed-rank test
   to decide which configuration is the best one instead of comparing the
   median difference.
                                                    (Manuel López-Ibáñez)

 * New functions `testing_fromlog()` and `testing_fromfile()` for independently
   executing the testing phase. The function `testing.main()` was removed as it
   is superseded by the new ones.
                                                    (Manuel López-Ibáñez)

 * New function `read_logfile()` to easily read the log file produced by irace.
                                                    (Manuel López-Ibáñez)

 * New function `printParameters()` that prints a parameters R object as a valid input text.
                                                    (Manuel López-Ibáñez)

 * `irace2pyimp` moved to its own R package.
                                                   (Manuel López-Ibáñez)

 * Generating the file `irace.Rdata` may be disabled by setting `logFile=""`.
                              (Manuel López-Ibáñez, reported by Johann Dreo)

 * `path_rel2abs()` and `checkParameters()` are now exported so that other
   packages may use them.
                                                    (Manuel López-Ibáñez)

 * `path_rel2abs()` also searches in system paths.     (Manuel López-Ibáñez)

 * `readConfigurationsFile()` will now detect duplicated configurations and
   error.                                          (Manuel López-Ibáñez)

 * The interface to functions `getFinalElites()`, `getConfigurationById()` and
   `getConfigurationByIteration()` has been simplified.

 * The package provides a `irace.sindef` file that may be used for building a
   standalone container of irace using Singularity. See the `README.md` file
   for instructions.                                (Contributed by Johann Dreo)

 * New example `examples/target-runner-python/target-runner-python-win.bat`
   contributed by Levi Ribeiro.

 * New helper script in `bin/parallel-irace-slurm` to launch `irace` in [SLURM](https://slurm.schedmd.com/) computer clu
                                                    (Manuel López-Ibáñez)

 * Rename `scenario.update.paths()` to `scenario_update_paths()`. The old name is deprecated. (Manuel López-Ibáñez)

## Fixes

 * Correctly handle clear out-performance cases despite strong bi-modality.
                                           (Reported by Nguyen Dang,
                                           fixed by Manuel López-Ibáñez)

 * Fix error when recovering from a parallel run on Windows.
                            (Manuel López-Ibáñez, reported by Tarek Gamal)

 * `testNbElites` now controls how many iteration elites are tested when
   `testIterationElites=1`. This is the documented behavior in the user guide.
                           (Manuel López-Ibáñez, reported by Marcelo de Souza)

 * Fixes to the Matlab example. (Manuel López-Ibáñez)
```

```
 * The default of `testType` is now set to `t-test` when capping is enabled.
                          (Manuel López-Ibáñez, reported by Jovana Radjenovic)

 * Fix various issues in the user guide.
                          (Manuel López-Ibáñez, reported by Jovana Radjenovic)

 * Remove duplicated elites.
                          (Manuel López-Ibáñez, reported by Federico Naldini)

 * Fix (#7): warnings with partial matched parameters.
                                  (Manuel López-Ibáñez, reported by Marc Becker)

 * Fix (#10): wrong assert with `elitist=0`.             (Manuel López-Ibáñez)

 * Fix (#12): irace can be run with [FastR](https://www.graalvm.org/22.1/docs/getting-started/#run-r).

 * Fix (#13): Maximum number configurations immediately rejected reached.
                                                              (Manuel López-Ibáñez)

 * Fix: when setting the scenario file in the command-line, `scenarioFile` was
   not set correctly. The correct scenario was used, however, the debug output
   and the value stored in the log / recovery file was wrong.
                         (Manuel López-Ibáñez, reported by Richard Schoonhoven)

 * With `sampleInstances = FALSE`, elitist irace does not change the order of
   instances already seen.  However, if you want to make sure that the order of
   the instances is enforced, you also need to set `elitistNewInstances=0`.

 * The function `irace.usage()` was removed. It was not really useful for R
   users as the same result can be obtained by calling
   `irace.cmdline("--help")`.
                                                              (Manuel López-Ibáñez)


# irace 3.4.1  (31/03/2020)

 * `NEWS` converted to markdown.

 * Fix CRAN error on Solaris.


# irace 3.4  (30/03/2020)

 * `irace2pyimp` function and executable (`irace2pyimp.exe` on Windows) to
   convert .Rdata files generated by irace to the input files required by the
   parameter importance analysis tool PyImp
   (https://github.com/automl/ParameterImportance).
                                             (Nguyen Dang, Manuel López-Ibáñez)

 * Initial configurations may also be provided directly in R using
   `scenario$initConfigurations`
                                                    (Manuel López-Ibáñez)

 * Rdata files are saved in version 2 to keep compatibility with older R
   versions.
                                                    (Manuel López-Ibáñez)
 * Fix invalid assert with ordered parameters:        (Leslie Pérez Cáceres)

    ```
    value >= 1L && value <= length(possibleValues) is not TRUE
    ```
```

```
 * The `irace` executable (`irace.exe` on Windows) is a compiled binary instead
   of a script. On Windows, `irace.exe` replaces `irace.bat`
                                                    (Manuel López-Ibáñez)

 * `inst/examples/Spear` contains the Spear (SAT solver) configuration scenario.
                                                    (Manuel López-Ibáñez)

 * Fixed bug when reporting minimum `maxTime` required.
                                             (Reported by Luciana Salete Buriol,
                                              fixed by Manuel López-Ibáñez)

 * Fixed bug detected by assert:

    ```R
    all(apply(!is.na(elite.data$experiments), 1, any)) is not TRUE
    ```

     (Reported by Maxim Buzdalov, fixed by Manuel López-Ibáñez)

irace 3.3 (26/04/2019)

 * Fix buggy test that breaks CRAN.   (Manuel López-Ibáñez)

 * Do not print "23:59:59" when wall-clock time is actually close to zero.
                                                         (Manuel López-Ibáñez)

irace 3.2 (24/04/2019)

 * Fix `irace --check --parallel 2` on Windows.   (Manuel López-Ibáñez)

 * Values of real-valued parameter are now printed with sufficient precision to
   satisfy `digits` (up to `digits=15`).
                                                         (Manuel López-Ibáñez)

 * It is possible to specify `boundMax` without capping.
                                      (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * `irace --check` will exit with code 1 if the check is unsuccessful
                                                         (Manuel López-Ibáñez)

 * Print where irace is installed with `--help`.   (Manuel López-Ibáñez)

 * irace will now complain if the output of `target-runner` or `target-evaluator`
   contains extra lines even if the first line of output is correct. This is to
   avoid parsing the wrong output. Unfortunately, this may break setups that
   relied on this behavior. The solution is to only print the output that irace
   expects.
                                                         (Manuel López-Ibáñez)

 * Completely re-implement `log` parameters to fix several bugs. Domains that
   contain zero or negative values are now rejected.
                                      (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * New option `aclib=` (`--aclib 1`) enables compatibility with the
   GenericWrapper4AC (https://github.com/automl/GenericWrapper4AC/) used by
   AClib (https://aclib.net/). This is EXPERIMENTAL. `--aclib 1` also sets
   digits to 15 for compatibility with AClib defaults.
                                                          (Manuel López-Ibáñez)

 * Fix printing of output when capping is enabled.
```

```
                                                        (Manuel López-Ibáñez)

 * `checkTargetFiles()` (`--check`) samples an instance unless
   `sampleInstances` is FALSE.                          (Manuel López-Ibáñez)

 * Fix symbol printed in elimination test.             (Manuel López-Ibáñez)

 * Use `dynGet()` to find `targetRunner` and `targetEvaluator`.
   As a result, we now require R >= 3.2.
                                                        (Manuel López-Ibáñez)

 * All tests now use `testthat`.                       (Manuel López-Ibáñez)

 * New function `scenario.update.paths()`              (Manuel López-Ibáñez)

 * Fix assert failure that may happen when `elitistNewInstances` is larger than
   `firstTest`. Reported by Jose Riveaux.              (Manuel López-Ibáñez)

 * Fix bug in `checkTargetFiles()` (`--check`) with capping.
                                                      (Leslie Pérez Cáceres)

 * Clarify a few errors/warnings when `maxTime > 0`.
                    (Manuel López-Ibáñez, suggested by Haroldo Gambini Santos)


# irace 3.1  (12/07/2018)

 * Use testthat for unit testing.                      (Manuel López-Ibáñez)

 * Allow instances to be a list of arbitrary R objects (`mlr` bugfix).
                                                        (Manuel López-Ibáñez)

# irace 3.0  (05/07/2018)

 * irace now supports adaptive capping for computation time minimization.
   The default value of the `testType` option is t-test when adaptive capping
   is enabled. Please see the user-guide for details.
                                  (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The package contains an `ablation()` function implementing the ablation
   method for parameter importance analysis by Fawcett and Hoos (2016).
                                   (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * New option `postselection` executes a post-selection race.
                                                      (Leslie Pérez Cáceres)

 * At the end of each race, if the race stops before evaluating all instances
   seen in previous races, then the best overall may be different than the best
   of the race. We now print the best overall (best-so-far). Elites evaluated
   on more instances are considered better than those evaluated on fewer.
                                  (Manuel López-Ibáñez, Leslie Pérez Cáceres)

 * Last active parameter values of numerical parameters (`i` and `r`) are carried
   by the sampling model. When a value must be assigned and the parameter was
   previously not active, the sampling is performed around the last value.
                                  (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * R help pages are now generated with Roxygen2.
                                  (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The user guide documents `--version`, `--help`, and `--check`.
                                                      (Manuel López-Ibáñez)
```

```
 * A return value of `Inf` from `targetRunner`/`targetEvaluation` results in
   the immediate rejection of the configuration without any further evaluation.
   This is useful for handling unreliable or broken configurations that should
   not stop irace.                                          (Manuel López-Ibáñez)

 * Numerical parameters may be sampled on a logarithmic scale using `i,log`
   or `r,log`.                                                (Alberto Franzin)

 * New `target-runner.bat` for Windows contributed by André de Souza Andrade.

 * Fixed all shell scripts calling functions before defining them, which is not
   portable.
                                                           (Manuel López-Ibáñez)

 * Fixed `--parallel` bug in Windows that resulted in
   `Error in checkForRemoteErrors(val)`.
                                                           (Manuel López-Ibáñez)

 * Improve error message when no training instances are given.
                                                           (Manuel López-Ibáñez)


# irace 2.4 (03/08/2017)

 * The output of irace now specifies in which order, if any, configurations are
   printed.
                             (Manuel López-Ibáñez, suggested by Markus Wagner)

 * Several fixes for handling paths in Windows.
                                                          (Manuel López-Ibáñez)

 * `readConfigurationsFile()` now has a `text=` argument, which allows reading
   configurations from a string.
                                                          (Manuel López-Ibáñez)

 * User-provided functions (targetRunner, targetEvaluator and
   repairConfiguration) and user-provided conditions for forbidden
   configurations are now byte-compiled when read, which should make their
   evaluation noticeably faster.
                                                           (Manuel López-Ibáñez)

 * The argument `'experiment'` passed to the R function `targetRunner` does not
   contain anymore an element `'extra.params'`. Similarly, the `'scenario'`
   structure does not contain anymore the elements `'instances.extra.params'` and
   `'testInstances.extra.params'`. Any instance-specific parameters values now
   form part of the character string that defines an instance and it is up to
   the user-defined `targetRunner` to parse them appropriately. These changes
   make no difference when targetRunner is an external script, or when
   instances and instance-specific parameter values are read from a file.
                                                           (Manuel López-Ibáñez)

# irace 2.3

 * Fix bug that will cause `iraceResults$experimentLog` to count calls to
   `targetEvaluator` as experiments, even if no call to `targetRunner` was
   performed. This does not affect the computation of the budget consumed and,
   thus, it does not affect the termination criteria of irace. The bug triggers
   an assertion that terminates irace, thus no run that was successful with
   version 2.2 is affected.
                                                          (Manuel López-Ibáñez)
```

```
# irace 2.2

 * Command-line parameters are printed to stdout (useful for future
   replications). (Manuel López-Ibáñez, suggested by Markus Wagner)

 * Users may provide a function to repair configurations before being
   evaluated. See the scenario variable repairConfiguration.
                                                    (Manuel López-Ibáñez)

 * The option `--sge-cluster` (`sgeCluster`) was removed and replaced by
   `--batchmode` (`batchmode`). It is now the responsibility of the target-runner
   to parse the output of the batch job submission command (e.g., `qsub` or
   `squeue`), and return just the job ID. Values supported are: "sge", "torque",
   "pbs" and "slurm".                                  (Manuel López-Ibáñez)

 * The option `--parallel` can now be combined with `--batchmode` to limit the
   number of jobs submitted by irace at once. This may be useful in batch
   clusters that have a small queue of jobs.
                                                       (Manuel López-Ibáñez)

 * New examples under `inst/examples/batchmode-cluster/`.
                                                     (Manuel López-Ibáñez)

 * It is now possible to include scenario definition files from other scenario
   files by using:

    ```R
    eval.parent(source("scenario-common.txt", chdir = TRUE, local = TRUE))
    ```

     This feature is VERY experimental and the syntax is likely to change in the
     future.   (Manuel López-Ibáñez)

 * Fix a bug that re-executed elite results under some circumstances.
   (Leslie Pérez Cáceres)

 * Restrict the number of maximum configurations per race to 1024.
   (Leslie Pérez Cáceres)

 * Do not warn if the last line in the instance file does not terminate with a
   newline. (Manuel López-Ibáñez)

 * Fix bug when `deterministic == 1`.
   (Manuel López-Ibáñez, Leslie Pérez Cáceres)

 * Update manual and vignette with details about the expected arguments and
   return value of `targetRunner` and `targetEvaluator`. (Manuel López-Ibáñez)

 * Many updates to the User Guide vignette. (Manuel López-Ibáñez)

 * Fix `\dontrun` example in `irace-package.Rd` (Manuel López-Ibáñez)

 * Fix bug: If testInstances contains duplicates, results of testing are not
   correctly saved in `iraceResults$testing$experiments` nor reported correctly
   at the end of a run. Now unique IDs of the form `1t, 2t, ...` are used for
   each testing instance. These IDs are used for the rownames of
   `iraceResults$testing$experiments` and the names of the
   `scenario$testInstances`
   and `iraceResults$testing$seeds` vectors. (Manuel López-Ibáñez)

 * Fix bug where irace keeps retrying the `target-runner` call even if it
   succeeds. (Manuel López-Ibáñez)
```

```
 * New command-line parameter
```
```
       --only-test FILE
```
   which just evaluates the configurations given in FILE on the testing
   instances defined by the scenario. Useful if you decide on the testing
   instances only after running irace.   (Manuel López-Ibáñez)

 * Bugfix: When using `maxTime != 0`, the number of experiments performed may be
   miscounted in some cases.   (Manuel López-Ibáñez)

irace 2.1

 * Fix CRAN errors in tests. (Manuel López-Ibáñez)

 * Avoid generating too many configurations at once if the initial time
   estimation is too small. (Manuel López-Ibáñez)

irace 2.0

 * Minimum R version is 2.15.

 * Elitist irace by default, it can be disabled with parameter `--elitist 0`.
   (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The parameter `--test-type` gains two additional values: (Manuel López-Ibáñez)

   - `t-test-bonferroni` (t-test with Bonferroni's correction for multiple
                       comparisons),
   - `t-test-holm` (t-test with Holm's correction for multiple comparisons)

 * MPI does not create log files with `--debug-level 0`.
   (Manuel López-Ibáñez)

 * For simplicity, the `parallel-irace-*` scripts do not use an auxiliary
   `tune-main` script. For customizing them, make a copy and edit them
   directly.
   (Manuel López-Ibáñez)

 * New parameters: (Manuel López-Ibáñez)
```
--target-runner-retries : Retry target-runner this many times in case of error.
```


 * We print diversity measures after evaluating on each instance:
   (Leslie Pérez Cáceres)

   - Kendall's W (also known as Kendall's coefficient of concordance) If 1,
     all candidates have ranked in the same order in all instances. If 0, the
     ranking of each candidate on each instance is essentially random.

             W = Friedman / (m * (k-1))

   - Spearman's rho: average (Spearman) correlation coefficient computed on the
     ranks of all pairs of raters. If there are no repeated data values, a
     perfect Spearman correlation of +1 or -1 occurs when each of the variables
     is a perfect monotone function of the other.

 * Many internal and external interfaces have changed. For example, now we

```
   consistently use 'scenario' to denote the settings passed to irace and
   'configuration' instead of 'candidate' to denote the parameter settings
   passed to the target algorithm. Other changes are:
```R
   parameters$boundary -> parameters$domain
   hookRun             -> targetRunner
   hookEvaluate        -> targetEvaluator
   tune-conf           -> scenario.txt
   instanceDir         -> trainInstancesDir
   instanceFile        -> trainInstancesFile
   testInstanceDir     -> testInstancesDir
   testInstanceFile    -> testInstancesFile
```

 * Minimal example of configuring a MATLAB program
   (thanks to Esteban Diaz Leiva)

 * Paths to files or directories given in the scenario file are relative to the
   scenario file (except for `--log-file`, which is an output file and it is
   relative to `--exec-dir`). Paths given in the command-line are relative to the
   current working directory. Given
```bash
      $ cat scenario/scenario.txt
      targetRunner <- "./target-runner"
      $ irace -s scenario/scenario.txt
```
   irace will search for `"./scenario/target-runner"`, but given
```bash
      $ irace -s scenario/scenario.txt --target-runner ./target-runner
```
   irace will search for `"./target-runner"`.   (Manuel López-Ibáñez)

 * New command-line wrapper for Windows installed at
   `system.file("bin/irace.bat", package="irace")`
   (thanks to Anthony Antoun)

 * Budget can be specified as maximum time (`maxTime`, `--max-time`) consumed by
   the target algorithm. See the documentation for the details about how this
   is handled.
   (Leslie Pérez Cáceres, Manuel López-Ibáñez)

irace 1.07

 * The best configurations found, either at the end or at each iteration of an
   irace run, can now be applied to a set of test instances different from the
   training instances. See options `testInstanceDir`, `testInstanceFile`,
   `testNbElites`, and `testIterationElites`. (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * The R interfaces of `hookRun`, `hookEvaluate` and `hookRunParallel` have changed.
   See `help(hook.run.default)` and `help(hook.evaluate.default)` for examples of
   the new interfaces.

 * Printing of race progress now reports the actual configuration and instance
   IDs, and numbers are printed in a more human-readable format.
   (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * Reduce memory use for very large values of `maxExperiments`.
   (Manuel López-Ibáñez, thanks to Federico Caselli for identifying the issue)

 * New option `--load-balancing` (`loadBalancing`) for disabling load-balancing
   when executing jobs in parallel. Load-balancing makes better use of
```

```
   computing resources, but increases communication overhead. If this overhead
   is large, disabling load-balancing may be faster.
   (Manuel López-Ibáñez, thanks to Federico Caselli for identifying the issue)

 * The option `--parallel` in Windows now uses load-balancing by default.
   (Manuel López-Ibáñez)

 * The wall-clock time after finishing each task is printed in the output.
   (Manuel López-Ibáñez, thanks to Federico Caselli for providing an initial
   patch)


# irace 1.06

 * Fix bug that could introduce spurious whitespace when printing the
   final configurations. (Manuel López-Ibáñez)

 * Fix bug if there are more initial candidates than needed for the
   first race. (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * New configuration options, mainly for R users:

    - `hookRunParallel`: Optional R function to provide custom
      parallelization of `hook.run`.

    - `hookRunData`: Optional data passed to `hookRun`. This is ignored by the
      default `hookRun` function, but it may be used by custom `hookRun` R
      functions to pass persistent data around.  (Manuel López-Ibáñez)

# irace 1.05

 * New option `--version`. (Manuel López-Ibáñez)

 * Terminate early if there is no sufficient budget to run irace with
   the given settings. (Manuel López-Ibáñez)

 * The option `--parallel` (without `--mpi`) now works under Windows.
   (Manuel López-Ibáñez, thanks to Pablo Valledor Pellicer for testing
   it)

 * Improved error handling when running under Rmpi. Now irace will
   terminate as soon as the master node detects at least one failed
   slave node. This avoids irace reporting two times the same error.
   Also, irace will print all the unique errors returned by all slaves
   and not just the first one.
   (Manuel López-Ibáñez)

 * Forbidden configurations may be specified in terms of constraints
   on their values. Forbidden configurations will never be evaluated by irace.
   See `--forbidden-file` and `inst/templates/forbidden.tmpl`.
   (Manuel López-Ibáñez)

 * New option `--recovery-file` (`recoveryFile`) allows resuming a
   previous irace run. (Leslie Pérez Cáceres)

 * The confidence level for the elimination test is now
   configurable with parameter `--confidence`. (Leslie Pérez Cáceres)

 * Much more robust handling of relative/absolute paths. Improved support
   for Windows. (Leslie Pérez Cáceres, Manuel López-Ibáñez)

 * Provide better error messages for incorrect parameter
```

```
   descriptions. (Manuel López-Ibáñez)
    Examples:
```
```
    x "" i (0, 0)       # lower and upper bounds are the same
    x "" r (1e-4, 5e-4) # given digits=2, ditto
    x "" i (-1, -2)     # lower bound must be smaller than upper bound
    x "" c ("a", "a")   # duplicated values
```
```
 * Print elapsed time for calls to hook-run if `debugLevel >=1`.
   (Manuel López-Ibáñez)

 * `examples/hook-run-python/hook-run`: A multi-purpose `hook-run` written
   in Python. (Franco Mascia)

 * Parallel mode in an SGE cluster (`--sge-cluster`) is more
   robust. (Manuel López-Ibáñez)

# irace 1.04

 * Replace obsolete package multicore by package parallel
   (requires R >= 2.14.0)

 * Use load-balancing (`mc.preschedule = FALSE`) in `mclapply`.

# irace 1.03

 * Use `reg.finalizer` to finish Rmpi properly without clobbering
   `.Last()`.

 * Remove uses of deprecated `as.real()`.

 * Nicer error handling in `readParameters()`.

 * Add hypervolume (multi-objective) example.

 * Fix several bugs in the computation of similar candidates.

# irace 1.02

 * More concise output.

 * The parameters `expName` and `expDescription` are now useless and they
   were removed.

 * Faster computation of similar candidates (Jeremie Dubois-Lacoste
   and Leslie Pérez Cáceres).

 * Fix bug when saving instances in `tunerResults$experiments`.

 * `irace.cmdline ("--help")` does not try to quit R anymore.

# irace 1.01

 * Fix bug caused by file.exists (and possibly other functions)
   not handling directory names with a trailing backslash or slash on
   Windows.

 * Fix bug using per-instance parameters (Leslie Pérez Cáceres).

 * Fix bug when reading initial candidates from a file.
```